

From Trigonometry to Conditional Descent Calculus
Functions, Change, and Certified Progress

Introduction

This text develops a coherent path from trigonometry to calculus and *Conditional Descent Calculus* (*CDC*), framework for processes that provably make progress via explicit potentials, precise guards, and stepwise certificates. Each lesson is self-contained: formal statements, a worked example, on-page references, and short problems. CDC tiles appear throughout: $\blacklozenge \Phi$ (potential), $\textcolor{red}{G}$ (guard), $\rightarrow \textcolor{blue}{U}$ (update), $\checkmark \textcolor{teal}{C}$ (certificate), $\square \textcolor{blue}{H}$ (halt). It should be noted that the symbols are merely for intuitive purposes. We do not propose new mathematics; we are proposing new methods to handle old problems.

Preface: Why CDC Now?

Motivation. Many classical methods succeed by tracking a nonnegative quantity that cannot decrease indefinitely. CDC abstracts this with guards and certificates, yielding local proofs that compose.

Historical threads. Euclid’s algorithm (strict descent on $\mathbb{N}$), bisection (halving), Lyapunov functions (monotone energy), potential-function methods in algorithms and optimization.

Alignment principles. Explicit Φ ; precise guards; certified steps; clear halting; composition laws; transparent error; parallel safety; minimal prereqs; auditability.

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Part I

Act I — Trig as Our Map

1. Angles and the Unit Circle

Context & Necessity

Angles are the most compact way to encode rotation and periodic behavior. Using *radians* turns geometry into arithmetic: arc length becomes a simple product, and the unit circle makes sin and cos precise as coordinates. This chapter fixes definitions and identities we will reuse in modeling, optimization, and later in CDC certificates where lengths, errors, and angular gaps serve as nonnegative potentials.

1.1 Definitions (symbol-first)

Definition 1.1 (Angle and radian). Given a circle of radius $r > 0$ and an arc of length $s \geq 0$, the *angle in radians* is

$$\theta = \frac{s}{r}.$$

One full revolution has $s = 2\pi r$, hence 2π radians.

Definition 1.2 (Unit circle, sine, and cosine). On the unit circle $x^2 + y^2 = 1$, let a point (x, y) be reached by counterclockwise rotation of angle θ from $(1, 0)$. Define

$$\cos \theta := x, \quad \sin \theta := y.$$

Lemma 1.3 (Arc length). *For any circle of radius r , the arc length and angle satisfy $s = r\theta$. On the unit circle ($r = 1$), $s = \theta$.*

Lemma 1.4 (Basic identities). *For all $\theta \in \mathbb{R}$:*

(a) **Pythagorean:** $\sin^2 \theta + \cos^2 \theta = 1$.

(b) **Parity:** $\sin(-\theta) = -\sin \theta, \quad \cos(-\theta) = \cos \theta$.

(c) **Periodicity:** $\sin(\theta + 2\pi) = \sin \theta, \quad \cos(\theta + 2\pi) = \cos \theta$.

1.2 Worked Example (applied, exact-to-approx)

Example 1.5 (Solar elevation from a shadow). A vertical pole of height $h = 2.0$ m casts a horizontal shadow of length $d = 3.5$ m. Find the solar elevation angle α above the horizon (in radians), then predict the shadow of a 10 m building at the same time.

Model. Right triangle with opposite = h , adjacent = d ,

$$\tan \alpha = \frac{h}{d} = \frac{2.0}{3.5} = \frac{4}{7}.$$

Thus $\alpha = \arctan(4/7) \approx 0.519$ rad ($\approx 29.74^\circ$).

Prediction. For building height $H = 10$, shadow length D satisfies $\tan \alpha = H/D \Rightarrow D = H/\tan \alpha = 10 / (4/7) = 17.5$ m.

Checks. (i) Units: $\tan$ is dimensionless; lengths scale linearly. (ii) Reasonableness: smaller angle $\Rightarrow$ longer shadow; 29.74° is modest, giving a longer D as expected.

Key Takeaways

- Radians: $\theta = s/r$ makes $s = r\theta$ algebraic and simplifies derivatives later.
- The unit circle defines $\sin, \cos$ as coordinates; identities follow from $x^2 + y^2 = 1$.
- Similar-triangle models (like shadows) convert geometry into function evaluation and inverse trig.

Try It Now

- 1) Ladder of length $L = 10$ m rests with its foot 6 m from a wall. Find the angle θ with the ground (radians). *Answer:* $\cos \theta = \frac{6}{10} = 0.6 \Rightarrow \theta = \arccos(0.6) \approx 0.9273$.
- 2) Unit circle coordinates: (a) $\theta = \frac{\pi}{6}$, (b) $\theta = \frac{5\pi}{6}$. *Answer:* (a) $(\cos, \sin) = (\sqrt{3}/2, 1/2)$; (b) $(-\sqrt{3}/2, 1/2)$.
- 3) Arc length: radius $r = 3$, central angle $\theta = 2.1$. *Answer:* $s = r\theta = 6.3$ (same units as r).

On-Page Reference

Radian/arc: $\theta = s/r$, so $s = r\theta$. **Unit circle:** $(\cos \theta, \sin \theta)$. **Pythagorean:** $\sin^2 \theta + \cos^2 \theta = 1$. **Even/odd:** $\cos$ even, $\sin$ odd. **Special angles:**

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0

Alignment Notes

Assumptions: Euclidean geometry, exact ratios before rounding. **Composes with:** identity algebra (next chapter), inverse functions and solving trig equations. **CDC connector:** angle/length errors are nonnegative potentials $\blacklozenge \Phi$; interval width on the unit circle is certifiably halved in bisection, giving logarithmic step bounds.

2. Waves and Periodicity

Context & Necessity

Periodic functions model rotation, oscillation, and any phenomenon that repeats. Writing them in *standard form* makes parameters interpretable: amplitude (size), period/frequency (repeat rate), phase (horizontal offset), and vertical shift (baseline). These parameters will later serve as explicit quantities (e.g., phase error) that we can bound or reduce.

2.1 Definitions and Standard Forms

Definition 2.1 (Standard sinusoid). A sinusoid in standard form is

$$y(x) = a \sin(b(x - c)) + d \quad \text{or} \quad y(x) = a \cos(b(x - c)) + d,$$

with parameters:

$$\text{amplitude } |a|, \quad \text{period } T = \frac{2\pi}{|b|}, \quad \text{phase shift } c, \quad \text{vertical shift } d.$$

Lemma 2.2 (Transform effects). For $y = a f(b(x - c)) + d$, where $f \in \{\sin, \cos\}$:

- (a) *Vertical scale:* $|a|$ stretches/compresses outputs; amplitude = $|a|$ for sinusoids.
- (b) *Horizontal scale:* b scales input; period $T = 2\pi/|b|$.
- (c) *Phase shift:* replace x by $(x - c)$ shifts right by c (left if $c < 0$).
- (d) *Vertical shift:* d moves the midline from 0 to d .

Lemma 2.3 (Sine–cosine equivalence). Every $a \cos(b(x - c)) + d$ can be written as $a \sin(b(x - c + \frac{\pi}{2b})) + d$, and conversely. Thus choice of $\sin$ vs. $\cos$ is a phase convention.

Lemma 2.4 (Amplitude–phase form). For real A, B , there exist unique $R \geq 0$ and φ with

$$A \sin x + B \cos x = R \sin(x + \varphi), \quad R = \sqrt{A^2 + B^2}, \quad \cos \varphi = \frac{A}{R}, \quad \sin \varphi = \frac{B}{R}.$$

2.2 Worked Example (parameter identification)

Example 2.5 (Fit a sinusoid from extrema and period). A signal $y(x)$ is known to be periodic with period $T = 24$. It attains a maximum value 30 at $x = 15$ and a minimum value 10 at $x = 3$. Find a standard-form model.

Midline and amplitude. Midline $d = \frac{30+10}{2} = 20$. Amplitude $|a| = \frac{30-10}{2} = 10$.

Frequency. $T = 24 \Rightarrow b = \frac{2\pi}{T} = \frac{\pi}{12}$.

Phase (choose cosine at a maximum). Cosine attains its maximum at zero phase. Since the maximum occurs at $x = 15$, set $x - c = 0 \Rightarrow c = 15$. A valid model is

$$y(x) = 20 + 10 \cos\left(\frac{\pi}{12}(x - 15)\right).$$

Check the minimum. Half a period after $x = 15$ is $x = 15 + T/2 = 27$; our given minimum is at $x = 3$, which is congruent to 27 (mod 24), so the placement is consistent: $y(3) = 20 + 10 \cos(\pi) = 10$.

Alternative sine form. Using sine with a quarter-period phase shift,

$$y(x) = 20 + 10 \sin\left(\frac{\pi}{12}\left(x - 15 + \frac{T}{4}\right)\right) = 20 + 10 \sin\left(\frac{\pi}{12}(x - 9)\right).$$

Both forms are equivalent.

Key Takeaways

- From $\{\max, \min\}$: $d = \frac{\max + \min}{2}$, $|a| = \frac{\max - \min}{2}$.
- From period T : $b = 2\pi/T$. Phase c aligns a known feature (e.g., a maximum) to zero phase.
- $A \sin x + B \cos x = R \sin(x + \varphi)$ consolidates mixed sin/cos into one phase.

Try It Now

1) Identify parameters. Given $y(x) = -3 \sin(2(x - \frac{\pi}{6})) + 5$, find amplitude, period, phase, shift.

Answer: $|a| = 3$, $T = \pi$, $c = \frac{\pi}{6}$ (right), $d = 5$; reflection across midline since $a < 0$.

2) Build a model from data. A periodic signal has period $T = 8$, midline $d = 2$, and minimum -1 at $x = 3$. Give a cosine model.

Answer: $|a| = 2 - (-1) = 3 \Rightarrow a = 3$ with a minimum requires a negated cosine at phase $x = 3$: $y(x) = 2 - 3 \cos(\frac{2\pi}{8}(x - 3)) = 2 - 3 \cos(\frac{\pi}{4}(x - 3))$.

3) Consolidate $A \sin + B \cos$. Write $y = 4 \sin x - 3 \cos x$ as $R \sin(x + \varphi)$.

Answer: $R = \sqrt{4^2 + (-3)^2} = 5$, $\cos \varphi = 4/5$, $\sin \varphi = -3/5$, so $y = 5 \sin(x + \varphi)$ with $\varphi = -\arctan(\frac{3}{4})$ (quadrant IV).

On-Page Reference

Standard form: $y = a \sin(b(x - c)) + d$ or $a \cos(b(x - c)) + d$. **Amplitude:** $|a|$.

Period: $T = \frac{2\pi}{|b|}$. **Phase shift:** c . **Midline:** $y = d$.

Extrema $\Rightarrow$ params: $d = \frac{\max + \min}{2}$, $|a| = \frac{\max - \min}{2}$.

$A \sin x + B \cos x$: $R = \sqrt{A^2 + B^2}$, $\cos \varphi = A/R$, $\sin \varphi = B/R$.

Shifts: $\sin(x \pm \alpha) = \sin x \cos \alpha \pm \cos x \sin \alpha$, $\cos(x \pm \alpha) = \cos x \cos \alpha \mp \sin x \sin \alpha$.

Alignment Notes

Assumptions: Real-valued, continuous sinusoids; period known or inferable. **Composes with:** identity algebra (next), inverse trig, solving phase equations. **CDC connector:** phase or amplitude mismatch can act as a nonnegative potential $\diamond \Phi$. Halving a phase interval by bisection yields logarithmic bounds to phase alignment.

3. Inverse Trig and Composition

Context & Necessity

Inverse trig functions recover angles from ratios. Getting domains, ranges, and compositions right prevents common algebraic errors and sets up exact triangle work, equation solving, and later CDC pages where angle gaps act as nonnegative potentials.

3.1 Definitions and Principal Values

Definition 3.1 (Inverse trigonometric functions).

$$\arcsin : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad \arccos : [-1, 1] \rightarrow [0, \pi], \quad \arctan : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

are the principal inverses of $\sin$, $\cos$, and $\tan$ on their standard restricted domains.

Lemma 3.2 (Compositions (principal values)). *For $x \in [-1, 1]$,*

$$\sin(\arcsin x) = x, \quad \cos(\arccos x) = x.$$

For $x \in \mathbb{R}$, $\tan(\arctan x) = x$.

Lemma 3.3 (Cross-compositions via right triangles). *For $x \in [-1, 1]$,*

$$\sin(\arccos x) = \sqrt{1 - x^2}, \quad \cos(\arcsin x) = \sqrt{1 - x^2}.$$

If a quadrant choice is specified (general angles), attach the appropriate sign.

3.2 Worked Example A (triangle reconstruction)

Example 3.4 (Recovering all trig values from a single ratio). Suppose $\cos \theta = \frac{3}{5}$ with θ in Quadrant I. Find $\sin \theta$, $\tan \theta$, and θ (principal).

Right-triangle model. Adjacent = 3, hypotenuse = 5 $\Rightarrow$ opposite = $\sqrt{5^2 - 3^2} = 4$.

$$\sin \theta = \frac{4}{5}, \quad \tan \theta = \frac{4}{3}, \quad \theta = \arccos(3/5) \text{ (numerically } \approx 0.9273 \text{ rad)}.$$

Check. Pythagorean identity: $(4/5)^2 + (3/5)^2 = 1$.

3.3 Worked Example B (solving a phase equation)

Example 3.5 (Solve $\sin(2x - \frac{\pi}{6}) = \frac{1}{2}$ on $[0, 2\pi)$). General solutions to $\sin u = \frac{1}{2}$ are $u = \frac{\pi}{6} + 2k\pi$ or $u = \frac{5\pi}{6} + 2k\pi$, $k \in \mathbb{Z}$.

Set $u = 2x - \frac{\pi}{6}$:

$$2x - \frac{\pi}{6} = \frac{\pi}{6} + 2k\pi \quad \text{or} \quad 2x - \frac{\pi}{6} = \frac{5\pi}{6} + 2k\pi.$$

Solve:

$$x = \frac{\pi}{6} + k\pi \quad \text{or} \quad x = \frac{\pi}{2} + k\pi.$$

List those in $[0, 2\pi)$:

$$x \in \left\{ \frac{\pi}{6}, \frac{7\pi}{6}, \frac{\pi}{2}, \frac{3\pi}{2} \right\}.$$

Key Takeaways

- Always declare domains/ranges: principal values prevent ambiguity.
- Use a right triangle to translate cross-compositions into radicals.
- For equations $\sin(ax+b) = c$, solve in the phase variable $u = ax+b$, then back-substitute and select solutions in the target interval.

Try It Now

1) Compute $\sin(\arccos \frac{5}{13})$.

Answer: $\sqrt{1 - (5/13)^2} = \frac{12}{13}$ (Quadrant I).

2) Solve $\cos(x - \frac{\pi}{4}) = \frac{\sqrt{2}}{2}$ on $[0, 2\pi)$.

Answer: $x - \frac{\pi}{4} = \pm \frac{\pi}{4} + 2k\pi \Rightarrow x \in \{0, \frac{\pi}{2}, 2\pi\}$; modulo 2π gives $\{0, \frac{\pi}{2}\}$.

3) Find all $x \in (-\pi, \pi]$ with $\tan x = -\sqrt{3}$.

Answer: $x = -\frac{\pi}{3}$ (principal); plus π -periodicity if a wider interval is requested.

On-Page Reference

Principal ranges: $\arcsin \in [-\pi/2, \pi/2]$, $\arccos \in [0, \pi]$, $\arctan \in (-\pi/2, \pi/2)$.

Cross-compositions: $\sin(\arccos x) = \sqrt{1 - x^2}$, $\cos(\arcsin x) = \sqrt{1 - x^2}$.

Phase method: let $u = ax + b$, solve in u , then divide by a and select interval solutions.

Alignment Notes

Assumptions: Principal branches unless a quadrant is specified. **Composes with:** parameter identification in sinusoids; inequality/angle bounds next chapter. **CDC connector:** phase residual $|ax + b - u^*|$ is a nonnegative potential $\blacklozenge \Phi$; bisection or halving on phase intervals yields logarithmic step bounds.

4. Inequalities, Bounds, and Scorekeeping

Context & Necessity

Inequalities are the grammar of conservative reasoning: they let us *bound* unknowns, control errors, and certify progress. In this text they serve three roles:

- (i) define nonnegative *potentials* (scores) we can monitor,
- (ii) propagate uncertainty through algebra and trig, and
- (iii) produce one-line *certificates* that a step is safe (drops add; factors multiply).

4.1 Foundational Laws

Lemma 4.1 (Order algebra). *For real numbers x, y, z :*

- (a) **Transitivity:** $x \leq y$ and $y \leq z \Rightarrow x \leq z$.
- (b) **Additive monotonicity:** $x \leq y \Rightarrow x + z \leq y + z$.
- (c) **Multiplicative monotonicity:** $0 \leq z \Rightarrow x \leq y \Rightarrow xz \leq yz$.
- (d) **Sign flip:** $z < 0 \Rightarrow x \leq y \Rightarrow xz \geq yz$.

Definition 4.2 (Absolute value). For $x \in \mathbb{R}$, $|x| = \max\{x, -x\}$, equivalently $|x| = \sqrt{x^2} \geq 0$.

Lemma 4.3 (Triangle and reverse triangle). *For all $x, y \in \mathbb{R}$,*

$$|x + y| \leq |x| + |y| \quad \text{and} \quad ||x| - |y|| \leq |x - y|.$$

Lemma 4.4 (AM-GM (two-term)). *For $a, b \geq 0$, $\frac{a+b}{2} \geq \sqrt{ab}$, with equality if $a = b$.*

Lemma 4.5 (Monotone functions preserve order). *If f is increasing on an interval and $x \leq y$ lies in it, then $f(x) \leq f(y)$. If f is decreasing, the inequality reverses.*

4.2 Bounding Trigonometric Quantities

Lemma 4.6 (Basic trig bounds). *For all $\theta \in \mathbb{R}$:*

$$-1 \leq \sin \theta \leq 1, \quad -1 \leq \cos \theta \leq 1, \quad |\sin \theta| \leq |\theta|.$$

Sketch. The first two follow from the unit-circle definition. For $|\sin \theta| \leq |\theta|$, one can compare arc length and vertical rise on the unit circle or use the series $|\sin \theta| \leq |\theta|$ for all θ . $\square$

Lemma 4.7 (Lipschitz bounds for sin/cos). *For all $a, b \in \mathbb{R}$,*

$$|\sin a - \sin b| \leq |a - b|, \quad |\cos a - \cos b| \leq |a - b|.$$

Proof. Use sum–difference identities: $\sin a - \sin b = 2 \cos\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$, so

$$|\sin a - \sin b| \leq 2 \cdot 1 \cdot \left| \sin\left(\frac{a-b}{2}\right) \right| \leq 2 \cdot \left| \frac{a-b}{2} \right| = |a - b|.$$

The cosine case is analogous via $\cos a - \cos b = -2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$. $\square$

4.3 Worked Example (error/control via inequalities)

Example 4.8 (Bounding phase error). Suppose a measured angle $\hat{\theta}$ approximates the true θ with $|\hat{\theta} - \theta| \leq \Delta$. Bound the output error in sin and cos:

$$|\sin \hat{\theta} - \sin \theta|, \quad |\cos \hat{\theta} - \cos \theta|.$$

Solution. By the Lipschitz bounds,

$$|\sin \hat{\theta} - \sin \theta| \leq |\hat{\theta} - \theta| \leq \Delta, \quad |\cos \hat{\theta} - \cos \theta| \leq \Delta.$$

Thus an angle error Δ produces at most Δ absolute error in sin or cos.

Interpretation (scorekeeping). If we take the *phase error* $\Phi := |\hat{\theta} - \theta| \geq 0$ as a potential, then any move that reduces phase error by δ reduces the output error by at least δ ; a direct "token style" certificate. If a procedure halves phase error, output error halves as well, a "factor style" certificate.

Key Takeaways

- Absolute value and triangle inequality are the default tools for conservative bounds.
- Trig is well conditioned: $|\sin a - \sin b| \leq |a - b|$ and $|\cos a - \cos b| \leq |a - b|$.
- A good potential $\Phi \geq 0$ aligns with the quantity you ultimately need to control.

Try It Now

1) Cosine bound. Show $|\cos a - \cos b| \leq |a - b|$.

Answer: Use $\cos a - \cos b = -2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$ and $|\sin| \leq 1$ and $|\sin u| \leq |u|$.

2) Algebraic bound. Prove $|x^2 - y^2| \leq (|x| + |y|)|x - y|$.

Answer: $|x^2 - y^2| = |(x - y)(x + y)| \leq |x - y|(|x| + |y|)$ by triangle.

3) Small-angle control. For $|x| \leq 0.1$, show $|\sin x| \leq 0.1$ and $|1 - \cos x| \leq \frac{x^2}{2} \leq 0.005$.

Answer: First from $|\sin x| \leq |x|$. Second via $1 - \cos x = 2 \sin^2(x/2) \leq 2(|x|/2)^2 = x^2/2$.

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Absolute value: $|x| = \max\{x, -x\} \geq 0$. **Triangle:** $|x + y| \leq |x| + |y|$. **Reverse triangle:** $||x| - |y|| \leq |x - y|$.

AM–GM: $\frac{a+b}{2} \geq \sqrt{ab}$ ($a, b \geq 0$). **Trig bounds:** $|\sin \theta| \leq 1$, $|\cos \theta| \leq 1$, $|\sin \theta| \leq |\theta|$.

Lipschitz: $|\sin a - \sin b| \leq |a - b|$, $|\cos a - \cos b| \leq |a - b|$.

Product bound: $|xy| \leq |x||y|$; extend via triangle to sums of products.

Alignment Notes

Assumptions: Real valued quantities; when using small angle forms, ensure $|x|$ is explicitly bounded. **Composes with:** parameter identification (in previous chapter) and CDC Step 1 certificates. **CDC connector:** Choosing Φ as an angle, length, or residual produces one line certificates: token style ($\Phi' \leq \Phi - \delta$) or factor style ($\Phi' \leq c\Phi$). Bounds here justify those certificates and make stopping thresholds auditable.

5. CDC Step 1: Tiles, Lemmas, and First Proofs

Context & Necessity

Many classical procedures terminate or converge because a nonnegative quantity cannot keep decreasing forever. *Conditional Descent Calculus (CDC)* isolates this idea: we act only under a **G** guard, perform an $\rightarrow \mathbf{U}$ update, and require a one-line $\checkmark \mathbf{C}$ certificate that a chosen $\blacklozenge \Phi$ potential decreases. The benefit is twofold:

- (i) *local proofs* per step
- (ii) *algebraic composition* (global bounds via adding drops or multiplying factors).

5.1 Definitions (tiles and valid steps)

Definition 5.1 (CDC tiles). A CDC configuration is a tuple (x, Φ, G, U, H) where x is the state, $\Phi : X \rightarrow \mathbb{R}_{\geq 0}$ is the potential, $G : X \rightarrow \{\mathbf{T}, \mathbf{F}\}$ is the guard, $U : X \rightarrow X$ is the update, and $H : X \rightarrow \{\mathbf{T}, \mathbf{F}\}$ is the halt predicate.

Definition 5.2 (Certified step). If $G(x)$ holds and one of the following is valid,

$$(\text{Token}) \quad \Phi(U(x)) \leq \Phi(x) - \delta \quad (\delta > 0), \quad \text{or} \quad (\text{Factor}) \quad \Phi(U(x)) \leq c \Phi(x) \quad (0 < c < 1),$$

then the update $x \leftarrow U(x)$ is *certified*. The process halts when $H(x)$ or when no guard is implemented.

Lemma 5.3 (Token Lemma). *If each legal step satisfies $\Phi' \leq \Phi - \delta$ with $\delta > 0$, then the process uses at most $\lceil \Phi(x_0)/\delta \rceil$ steps.*

Proof. After k steps, $\Phi(x_k) \leq \Phi(x_0) - k\delta \geq 0$, so $k \leq \Phi(x_0)/\delta$. □

Lemma 5.4 (Halving Lemma). *If each legal step satisfies $\Phi' \leq c \Phi$ with $0 < c < 1$, then $\Phi(x_k) \leq c^k \Phi(x_0)$. To obtain $\Phi \leq \varepsilon$, it suffices that $k > \log_{1/c}(\Phi(x_0)/\varepsilon)$.*

Theorem 5.5 (Sequential composition). *If step A guarantees $\Phi' \leq \Phi - \delta_A$ and step B guarantees $\Phi'' \leq \Phi' - \delta_B$, then $\Phi'' \leq \Phi - (\delta_A + \delta_B)$. If A and B are factor steps with constants $c_A, c_B \in (0, 1)$, then $\Phi'' \leq (c_B c_A) \Phi$.*

Proof. Substitute the first inequality into the second; for factors, multiply. □

5.2 Worked Example A (discrete): Binary search on $[1, N]$

Example 5.6 (Halving a candidate set). We wish to identify a secret integer in $\{1, \dots, N\}$ by yes/no queries that halve the candidate set.

◆ Φ Φ = current set size; G $\Phi > 1$; $\rightarrow \text{U}$ keep the half consistent with the answer; $\checkmark \text{C}$ $\Phi' \leq \frac{1}{2}\Phi$; $\square \text{H}$ $\Phi = 1$.

By Theorem 5.4, after k questions $\Phi_k \leq N/2^k$. Hence $k \geq \lceil \log_2 N \rceil$ suffices to isolate the secret.

5.3 Worked Example B (combinatorial): Sorting by fixing inversions

Definition 5.7 (Inversions). For a list $a_1, \dots, a_n$, an *inversion* is a pair (i, j) with $i < j$ and $a_i > a_j$. Let $\text{Inv}(a)$ be the number of inversions.

Example 5.8 (Adjacent swap reduces inversions). Consider the rule: if an adjacent pair is inverted, swap it.

◆ Φ $\Phi = \text{Inv}(a) \geq 0$; G there exists i with $a_i > a_{i+1}$; $\rightarrow \text{U}$ swap (a_i, a_{i+1}) ; $\checkmark \text{C}$ $\Phi' \leq \Phi - 1$ (the swapped inverted pair is removed, others do not increase); $\square \text{H}$ no adjacent inversion $\Rightarrow$ list sorted.

By Theorem 5.3, at most $\text{Inv}(a^{(0)})$ swaps are required.

5.4 Worked Example C (number-theoretic): Euclid's algorithm

Example 5.9 (Strict descent on $\mathbb{N}$). State (a, b) with $a \geq b > 0$; update $(a, b) \mapsto (b, a \bmod b)$.

◆ Φ $\Phi = b$; G $b \neq 0$; $\rightarrow \text{U}$ $(b, a \bmod b)$; $\checkmark \text{C}$ $a \bmod b < b \Rightarrow \Phi' < \Phi$; $\square \text{H}$ $b = 0$.

By Theorem 5.3, steps $\leq b_0$ (coarse). Sharper: $O(\log a_0)$ via Fibonacci bounds.

Key Takeaways

- Declare $\Phi \geq 0$ first; design G and U so a one-line certificate is easy.
- Global effort bounds follow from local guarantees: drops add; factors multiply.
- Choose Φ to match the goal (size of a candidate set, count of “bad” pairs, current modulus).

Try It Now

1) Token bound. If $\Phi_0 = 91$ and each step drops by at least $\delta = 4$, what is the worst-case step count?

Answer: $\lceil 91/4 \rceil = 23$.

2) Halving bound. Starting at $\Phi_0 = 100$, factor step $c = \frac{1}{2}$, target $\varepsilon = 0.8$. Minimal k ?

Answer: $k > \log_2(100/0.8) = \log_2(125) \approx 6.97 \Rightarrow 7$.

3) Inversions. For $[3, 2, 1, 4]$, compute Inv and bound the swaps needed by the adjacent-swap rule.

Answer: Inversions $\{(3, 2), (3, 1), (2, 1)\} \Rightarrow \Phi_0 = 3; \leq 3$ swaps.

4) Compose. One token step with $\delta = 5$ followed by a factor step with $c = 0.6$. Give a bound on Φ'' in terms of Φ .

Answer: $\Phi' \leq \Phi - 5$, then $\Phi'' \leq 0.6(\Phi - 5) = 0.6\Phi - 3$.

On-Page Reference

Certificates: (Token) $\Phi' \leq \Phi - \delta$ ($\delta > 0$); (Factor) $\Phi' \leq c\Phi$ ($0 < c < 1$). **Logs:** $c^k \leq \varepsilon/\Phi_0 \iff k \geq \log_{1/c}(\Phi_0/\varepsilon)$.

Inversions: count of $(i < j, a_i > a_j)$. Adjacent swap $\Rightarrow \Delta\Phi \leq -1$.

Composition: drops add; factors multiply; mixed gives affine bound $c(\Phi - \delta)$.

Common Pitfalls

- *Non-total guards:* ensure at least one rule is enabled unless H holds.
- *Bad potential:* if Φ can increase or go negative, guarantees collapse.
- *Hidden interactions:* parallel steps that touch the same part of the state can invalidate certificates (handled in later chapters).

Alignment Notes

Guarantees: termination (Token) or accuracy (Factor). **Compose:** Theorem 5.5. **Next:** choice/parallel safety and invariants (CDC Step 2).

Part II

Act II — Functions Grow Up

6. Exponentials and Logarithms

Context & Necessity

Exponentials model *multiplicative* change: growth/decay at a rate proportional to the current amount (populations, radioactivity, interest, repeated percentage updates). Logarithms invert exponentials, turning multiplicative relations into additive ones, and time-to-target problems into linear solves. For CDC, a fixed per-step factor is a *factor certificate*: $\Phi' \leq c\Phi$ with $0 < c < 1$ for decay (or $c > 1$ for growth when Φ measures the distance to a cap).

6.1 Definitions and Core Laws

Definition 6.1 (Exponential). For a base $a > 0$, $a \neq 1$, define a^x for real x by $a^x = e^{x \ln a}$. The *natural exponential* e^x uses the base $e \approx 2.71828$.

Definition 6.2 (Logarithm (inverse)). For $a > 0$, $a \neq 1$, the logarithm $\log_a x$ is the inverse of a^x : $a^{\log_a x} = x$ for $x > 0$. The natural logarithm is $\ln x = \log_e x$.

Lemma 6.3 (Laws (for $x, y > 0$, $a, b > 0$, $a, b \neq 1$)).

$$\begin{aligned} a^{x+y} &= a^x a^y, & a^{xy} &= (a^x)^y, & a^{-x} &= \frac{1}{a^x}, \\ \log_a(xy) &= \log_a x + \log_a y, & \log_a\left(\frac{x}{y}\right) &= \log_a x - \log_a y, & \log_a(x^r) &= r \log_a x, \\ \log_a x &= \frac{\ln x}{\ln a} \text{ (change of base).} \end{aligned}$$

Lemma 6.4 (Calculus rules).

$$\begin{aligned} \frac{d}{dx} e^x &= e^x, & \frac{d}{dx} a^x &= (\ln a) a^x, & \frac{d}{dx} \ln x &= \frac{1}{x} \quad (x > 0). \\ \int e^x dx &= e^x + C, & \int a^x dx &= \frac{a^x}{\ln a} + C, & \int \frac{1}{x} dx &= \ln|x| + C. \end{aligned}$$

Definition 6.5 (Continuous-rate model). A quantity following proportional change satisfies $y'(t) = k y(t)$ with solution $y(t) = y_0 e^{kt}$. Here k is the *continuous* growth ($k > 0$) or decay ($k < 0$) rate, doubling time $T_2 = \frac{\ln 2}{k}$, half-life $T_{1/2} = \frac{\ln 2}{|k|}$.

6.2 Worked Example A (rate $\rightarrow$ model $\rightarrow$ time)

Example 6.6 (From doubling time to rate, then to when it triples). A culture doubles every 8 hours. Find k and the time to triple.

Rate. $T_2 = 8$ h, so $k = \frac{\ln 2}{8} \approx 0.08664 \text{ h}^{-1}$.

Tripling time. Solve $3 = e^{kt}$: $t = \frac{\ln 3}{k} = \frac{8 \ln 3}{\ln 2} \approx 12.66$ h.

6.3 Worked Example B (half-life to threshold)

Example 6.7 (How long until only 10% remains?). A substance has half-life 30 minutes. Solve $y(t) = y_0 e^{kt}$ for t when $y/y_0 = 0.1$.

Rate. $k = -\frac{\ln 2}{0.5 \text{ h}} = -2 \ln 2 \approx -1.3863 \text{ h}^{-1}$.

Time. $0.1 = e^{kt} \Rightarrow t = \frac{\ln 0.1}{k} \approx 1.66 \text{ h} \approx 99.5 \text{ min}$.

6.4 Worked Example C (solving an exponential equation)

Example 6.8 (Solve for time from a measurement). Given $7e^{0.6t} = 20$, solve for t .

$$e^{0.6t} = \frac{20}{7} \Rightarrow 0.6t = \ln\left(\frac{20}{7}\right) \Rightarrow t = \frac{1}{0.6} \ln\left(\frac{20}{7}\right) \approx 1.73.$$

6.5 Worked Example D (calculus with exp/log)

Example 6.9 (Differentiate and integrate). (a) $f(x) = 2^x + \ln(3x^2)$:

$$f'(x) = (\ln 2) 2^x + \frac{1}{3x^2} \cdot 6x = (\ln 2) 2^x + \frac{2}{x}.$$

(b) $\int (5e^{5x} + 1/x) dx = e^{5x} + \ln|x| + C.$

Key Takeaways

- Exponentials model constant *relative* change; logarithms invert them and linearize multiplicative relations.
- The continuous model $y(t) = y_0 e^{kt}$ encodes doubling time $T_2 = \ln 2/k$ and half-life $T_{1/2} = \ln 2/|k|$.
- Derivatives: $(e^x)' = e^x$, $(a^x)' = (\ln a)a^x$, $(\ln x)' = 1/x$; integrals mirror these.

Try It Now

1) Rate from a percentage. If a quantity grows 8% per year continuously, what is k and the doubling time?

Answer: $k = \ln(1.08) \approx 0.07696 \text{ yr}^{-1}$, $T_2 = \ln 2/k \approx 9.01 \text{ yr}$.

2) Solve for t . $12 \cdot 3^{0.2t} = 30$.

Answer: $0.2t = \log_3(30/12) = \log_3(2.5)$, so $t = 5 \log_3(2.5) = \frac{5 \ln 2.5}{\ln 3} \approx 3.64$.

3) Half-life. Half-life 5 days. Time for $y/y_0 = 1/6$?

Answer: $k = -\ln 2/5$, so $t = \ln(1/6)/k = \frac{5 \ln 6}{\ln 2} \approx 12.9 \text{ days}$.

4) Differentiate. $g(x) = a^x \ln x$ ($a > 1$).

Answer: $g'(x) = (\ln a) a^x \ln x + \frac{a^x}{x} = a^x \left((\ln a) \ln x + \frac{1}{x} \right)$.

On-Page Reference

Inverse: $a^{\log_a x} = x$, $\ln(e^x) = x$, $e^{\ln x} = x$.

Change of base: $\log_a x = \frac{\ln x}{\ln a}$. **Rates:** $y(t) = y_0 e^{kt}$; $T_2 = \ln 2/k$; $T_{1/2} = \ln 2/|k|$.

Derivatives: $(e^x)' = e^x$, $(a^x)' = (\ln a) a^x$, $(\ln x)' = 1/x$.

Integrals: $\int e^x = e^x$, $\int a^x = \frac{a^x}{\ln a}$, $\int dx/x = \ln |x| + C$.

Alignment Notes

Assumptions: $\ln x$ only for $x > 0$; rates k constant over the modeled window.

Composes with: series (Taylor for e^x), limits (definitions of e), optimization (log-transform).

CDC connector: Treat the amount (or error) as $\diamond \Phi$. A fixed per-step multiplicative change gives a *factor* certificate $\Phi' \leq c\Phi$. Logs linearize: $\log \Phi_k = \log \Phi_0 + k \log c$, so the step bound $k \geq \frac{\log(\Phi_0/\varepsilon)}{\log(1/c)}$ is immediate.

7. Polynomials and Factorization

Context & Necessity

Polynomials are the canonical finite models for smooth behavior: closed under addition/multiplication, with roots that encode structure. For solving equations and bounding behavior we use three tools:

- (i) *Factor/Remainder Theorems* (evaluate to divide, divide to factor),
- (ii) *Rational Root Test* (candidate screening when coefficients are integers),
- (iii) *Multiplicity/End Behavior* (qualitative shape).

These support CDC style arguments when a sign change certifies an interval containing a root and halving gives a logarithmic step bound.

7.1 Definitions and Basic Facts

Definition 7.1 (Polynomial, degree, leading coefficient). A polynomial over $\mathbb{R}$ is $p(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_i \in \mathbb{R}$, $a_n \neq 0$. The *degree* is n , the *leading coefficient* is a_n .

Definition 7.2 (Multiplicity). If $(x - r)^m$ divides p but $(x - r)^{m+1}$ does not, then r is a *root of multiplicity* m . Even multiplicity $\Rightarrow$ the graph *touches* the axis at r ; odd $\Rightarrow$ it *crosses*.

Lemma 7.3 (Factor and Remainder Theorems). *For any real number r and polynomial p :*

$$p(x) = (x - r)q(x) + p(r).$$

In particular, $p(r) = 0 \iff (x - r)$ divides $p(x)$.

Lemma 7.4 (Rational Root Test (integer coefficients)). *Let $p(x) = a_n x^n + \cdots + a_0$ with $a_i \in \mathbb{Z}$ and $a_0 \neq 0$. Any rational root $r = \frac{u}{v}$ in lowest terms satisfies*

$$u \mid a_0, \quad v \mid a_n.$$

Thus candidates are $\pm$ (factors of a_0)/(factors of a_n).

Lemma 7.5 (End behavior). *If n is even, $a_n > 0$ then $p(x) \rightarrow +\infty$ as $x \rightarrow \pm\infty$ (and $\rightarrow -\infty$ if $a_n < 0$). If n is odd, $a_n > 0$ then $p(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $\rightarrow +\infty$ as $x \rightarrow +\infty$ (reverse if $a_n < 0$).*

7.2 Worked Example A (factor a cubic via a rational root)

Example 7.6 (Rational Root Test $\rightarrow$ synthetic division $\rightarrow$ full factorization). Factor $p(x) = 2x^3 - 3x^2 - 11x + 6$ over $\mathbb{R}$.

Candidates. $a_0 = 6$ (factors $\pm 1, \pm 2, \pm 3, \pm 6$), $a_3 = 2$ (denominators $\pm 1, \pm 2$). Test $x = 3$: $p(3) = 2 \cdot 27 - 3 \cdot 9 - 11 \cdot 3 + 6 = 54 - 27 - 33 + 6 = 0$.

Divide by $(x-3)$. Synthetic division on coefficients $(2, -3, -11, 6)$ by 3 yields the quotient $2x^2 + 3x - 2$.

Finish. $2x^2 + 3x - 2 = (2x - 1)(x + 2)$, so

$$\boxed{p(x) = (x - 3)(2x - 1)(x + 2)}.$$

Check. Three real roots: $3, \frac{1}{2}, -2$; leading coefficient 2 matches.

7.3 Worked Example B (construct from roots and multiplicities)

Example 7.7 (Build a polynomial with specified zeros). Find the least-degree polynomial with real coefficients, leading coefficient 3, having a double root at $x = 1$ and a simple root at $x = -2$. Evaluate $p(0)$.

Form. $(x - 1)^2(x + 2)$ encodes the multiplicities; scale by 3:

$$p(x) = 3(x - 1)^2(x + 2).$$

Value at 0. $p(0) = 3(1)^2(2) = \boxed{6}$.

7.4 Worked Example C (remainder without dividing)

Example 7.8 (Remainder Theorem in practice). Let $p(x) = x^4 - 2x^3 + x - 7$. Find the remainder on division by $(x + 1)$ and by $(x - 2)$.

By Remainder Theorem. $r_{x+1} = p(-1) = 1 + 2 - 1 - 7 = \boxed{-5}$, $r_{x-2} = p(2) = 16 - 16 + 2 - 7 = \boxed{-5}$.

Key Takeaways

- Use *Remainder Theorem* to evaluate division by $(x-r)$ instantly; zero remainder certifies a factor.
- For integer coefficients, *Rational Root Test* gives a finite candidate list.
- Multiplicity encodes crossing vs. touching and helps sketch graphs quickly alongside end behavior.

Try It Now

1) Factor with rational roots. Factor $x^3 - 4x^2 - 7x + 10$.

Answer: Test $x = 1$: $1 - 4 - 7 + 10 = 0 \Rightarrow (x - 1)$; quotient $x^2 - 3x - 10 = (x - 5)(x + 2)$. So $(x - 1)(x - 5)(x + 2)$.

2) Remainders quickly. For $p(x) = 2x^3 + x^2 - 5x + 4$, compute remainders mod $(x - 1)$ and $(x + 2)$.

Answer: $p(1) = 2 + 1 - 5 + 4 = 2$; $p(-2) = -16 + 4 + 10 + 4 = 2$.

3) Multiplicity and sign. Describe the x -intercept behavior of $f(x) = (x - 2)^2(x + 1)$.

Answer: Touches (does not cross) at $x = 2$ (even mult. 2); crosses at $x = -1$ (odd mult.).

On-Page Reference

Factor/Remainder: $p(x) = (x - r)q(x) + p(r)$; $p(r) = 0 \iff (x - r) \mid p(x)$.

Rational Root Test: candidates $\pm \text{factors}(a_0)/\text{factors}(a_n)$ for integer coefficients.

Vieta (quick cases): For $ax^2 + bx + c$, $r_1 + r_2 = -b/a$, $r_1 r_2 = c/a$. For $ax^3 + bx^2 + cx + d$, $r_1 + r_2 + r_3 = -b/a$, $\sum r_i r_j = c/a$, $r_1 r_2 r_3 = -d/a$.

End behavior: parity of degree + sign of a_n governs tails.

Multiplicity: even $\Rightarrow$ touch; odd $\Rightarrow$ cross.

Alignment Notes

Assumptions: Real coefficients unless stated; Rational Root Test requires integer coefficients.

Composes with: inequality bounds (root bracketing), sinusoid fitting (polynomial envelopes).

CDC connector: If $p(L) \cdot p(R) < 0$, the interval $[L, R]$ contains a real root. Take $\blacklozenge \Phi$ as interval width and apply halving (bisection); the sign test is the guard $\textcolor{red}{G}$, halving is the update $\rightarrow \textcolor{blue}{U}$, the factor certificate $\checkmark \textcolor{teal}{C}$ is $\Phi' \leq \frac{1}{2}\Phi$, and $\square \textcolor{blue}{H}$ is width $\leq \varepsilon$.

8. Rational, Piecewise, and Absolute Functions

Context & Necessity

These families introduce domains with exclusions and rule based definitions. Analyzing *where* a formula applies (and how it behaves near boundaries) is core to rigorous modeling. For CDC later on, domains become **G** guards, and well chosen nonnegative quantities (distance to a break-point, interval width, or sign gaps) become **◆** **Φ** potentials with one-line certificates.

8.1 Definitions and Core Facts

Definition 8.1 (Rational function). A rational function is $f(x) = \frac{p(x)}{q(x)}$ with polynomials p, q and $q \not\equiv 0$. Its *domain* is $\{x \in \mathbb{R} : q(x) \neq 0\}$.

Lemma 8.2 (Asymptotes and degree test). Let $\deg p = n$, $\deg q = m$.

- (a) If a factor $(x - r)$ cancels between p and q , $x = r$ is a removable hole, not an asymptote.
- (b) If $q(r) = 0$ after cancellation, $x = r$ is a vertical asymptote.
- (c) If $n < m$, then $y = 0$ is a horizontal asymptote.
- (d) If $n = m$, then $y = \frac{\text{lead}(p)}{\text{lead}(q)}$ is a horizontal asymptote.
- (e) If $n = m + 1$, long division gives an oblique (slant) asymptote $y = ax + b$.

Definition 8.3 (Piecewise function and continuity at a break-point). A function defined by cases $f(x) = \begin{cases} f_1(x) & x < c \\ f_2(x) & x \geq c \end{cases}$ is *continuous at c* if both one-sided limits exist and equal $f(c)$: $\lim_{x \uparrow c} f_1(x) = \lim_{x \downarrow c} f_2(x) = f(c)$.

Definition 8.4 (Absolute value (piecewise form)). $|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$. Thus $|ax + b|$ is linear on each side of $x = -b/a$, forming a V-shape with its vertex at that point.

8.2 Worked Example A (rational analysis end-to-end)

Example 8.5 (Asymptotes, domain, intercepts, sketch). Analyze $f(x) = \frac{x^2 - 1}{x - 2}$.

Domain. $x \neq 2$ (no cancellation with numerator).

Long division. $x^2 - 1 = (x - 2)(x + 2) + 3 \Rightarrow f(x) = x + 2 + \frac{3}{x - 2}$.

Asymptotes. Vertical: $x = 2$ (denominator = 0). Oblique: $y = x + 2$ (degree 2 = 1 + 1).

Intercepts. x -intercepts solve $x^2 - 1 = 0 \Rightarrow x = \pm 1$. y -intercept: $f(0) = (-1)/(-2) = \frac{1}{2}$.

Behavior near $x = 2$. From the right, $\frac{3}{x - 2} \rightarrow +\infty$; from the left, $\rightarrow -\infty$.

8.3 Worked Example B (make a piecewise function continuous)

Example 8.6 (Choose a parameter for continuity). Define

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1, \\ A, & x = 1. \end{cases}$$

Find A so that f is continuous at $x = 1$.

Simplify off the hole. For $x \neq 1$, $(x^2 - 1)/(x - 1) = x + 1$.

Match the limit. $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x + 1) = 2$. Set $A = 2$ for continuity.

8.4 Worked Example C (absolute value equations and inequalities)

Example 8.7 (Solve $|2x - 3| \leq 5$ and express the solution interval). By piecewise reasoning or the standard rule, $|u| \leq k \iff -k \leq u \leq k$ for $k \geq 0$. Here $u = 2x - 3$, $k = 5$:

$$-5 \leq 2x - 3 \leq 5 \implies -1 \leq 2x \leq 8 \implies \boxed{-\frac{1}{2} \leq x \leq 4}.$$

Key Takeaways

- Always cancel common factors before declaring vertical asymptotes; cancellations produce removable holes.
- Degree comparison gives end behavior: $(n < m) \Rightarrow y = 0$; $(n = m) \Rightarrow$ constant ratio; $(n = m + 1) \Rightarrow$ slant.
- Piecewise continuity at a break-point requires left limit = right limit = value.
- Absolute values become linear on each side of the vertex; inequalities translate to interval statements.

Try It Now

1) Rational sketch quick-pass. For $g(x) = \frac{x^2 - 4}{x^2 - 1}$, find domain, vertical/horizontal asymptotes, and any holes.

Answer: Factor: $g = \frac{(x-2)(x+2)}{(x-1)(x+1)}$ (no cancellation). Domain: $x \neq \pm 1$. $\deg p = \deg q = 2 \Rightarrow$ horizontal $y = 1$. Vertical asymptotes $x = \pm 1$. No holes.

2) Continuity parameter. Let $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ A, & x = 0 \end{cases}$. Choose A for continuity (use the standard limit).

Answer: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \Rightarrow A = 1$.

3) Absolute value equation. Solve $|x + 2| = 3$ and express the solution set.

Answer: $x + 2 = 3$ or $x + 2 = -3 \Rightarrow x = 1$ or $x = -5$. Solution $\{-5, 1\}$.

On-Page Reference

Rational checklist: Cancel common factors $\rightarrow$ identify holes; remaining zeros of $q \rightarrow$ vertical asymptotes.

Degree test: $n < m \Rightarrow y = 0$; $n = m \Rightarrow y =$ leading-coefficient ratio; $n = m + 1 \Rightarrow$ slant from long division.

Piecewise continuity: $\lim_{x \uparrow c} f = \lim_{x \downarrow c} f = f(c)$.

Absolute value: $|u| \leq k \iff -k \leq u \leq k$; $|u| < k$ gives an open interval; $|u| \geq k$ gives two rays.

Sign charts: factor numerators/denominators, mark zeros and exclusions, test intervals.

Alignment Notes

Assumptions: Real coefficients, explicit cancellation before asymptote claims; standard limit facts used where noted. **Composes with:** inequalities (sign charts), inverse trig (domain-aware equations), and later optimization. **CDC connector:** Domain restrictions become **G** guards; removable holes suggest invariant maintenance. When bracketing a root between sign changes, $\diamond \Phi$ as interval width with halving $\rightarrow$ **U** gives a factor certificate $\checkmark \mathbf{C}$ and explicit step bounds.

9. Sequences and Series

Context & Necessity

Sequences encode stepwise processes; series encode cumulative effect. Geometric decay is the algebra behind factor-style certificates: if each step scales a quantity by $c \in (0, 1)$, then the error after k steps is c^k times the start, and the *remaining tail* is bounded by a geometric sum.

9.1 Definitions and Core Results

Definition 9.1 (Sequence, convergence). A *sequence* is a function $a : \mathbb{N} \rightarrow \mathbb{R}$, written $(a_n)_{n \geq 0}$. We say $a_n \rightarrow L$ if for every $\varepsilon > 0$ there exists N such that $n \geq N \Rightarrow |a_n - L| < \varepsilon$.

Definition 9.2 (Series, partial sums). Given (a_n) , the *series* $\sum_{n=0}^{\infty} a_n$ has partial sums $S_N = \sum_{n=0}^N a_n$. If S_N converges to S , we write $\sum_{n=0}^{\infty} a_n = S$.

Lemma 9.3 (Geometric sequence and series). Let $r \in \mathbb{R}$.

(a) $a_n = a_0 r^n$.

(b) For $r \neq 1$, $\sum_{n=0}^N ar^n = a \frac{1 - r^{N+1}}{1 - r}$.

(c) If $|r| < 1$, then $\sum_{n=0}^{\infty} ar^n = \frac{a}{1 - r}$ and the tail $R_N := \sum_{n=N+1}^{\infty} ar^n = a \frac{r^{N+1}}{1 - r}$.

Lemma 9.4 (Monotone convergence (bounded)). If (a_n) is increasing and bounded above, then a_n converges. If it is decreasing and bounded below, it converges.

9.2 Worked Example A (finite geometric model)

Example 9.5 (Savings plan). Deposit \$200 at the end of each month into an account yielding 0.6% per month. After N months, how much is accumulated?

Model. Each deposit compounds for a different number of months. Let $r = 1.006$. Then

$$S_N = 200 (1 + r + r^2 + \cdots + r^{N-1}) = 200 \frac{r^N - 1}{r - 1}.$$

Solve $S_N \geq 10,000$ for N . Rearranging, $r^N \geq 1 + \frac{(r - 1)S_N}{200}$. Take $\ln$ to isolate N .

9.3 Worked Example B (tail/error bound)

Example 9.6 (Choose N so that the remainder is $< \varepsilon$). Approximate $S = \sum_{n=0}^{\infty} 3\left(\frac{1}{2}\right)^n$ by S_N so that the tail $R_N < 10^{-3}$.

Tail bound. With $a = 3$, $r = \frac{1}{2}$, $R_N = a \frac{r^{N+1}}{1-r} = 3 \cdot \frac{(1/2)^{N+1}}{1/2} = 3 \cdot 2 \cdot (1/2)^{N+1} = 3 \cdot (1/2)^N$. We need $3 \cdot 2^{-N} < 10^{-3} \Rightarrow 2^N > 3 \cdot 10^3$. Thus $N > \log_2(3000) \approx 11.55$, so $\boxed{N = 12}$ suffices.

9.4 Worked Example C (first-order linear recurrence)

Example 9.7 (Closed form and limit). Let $x_{k+1} = 0.8x_k + 5$, $x_0 = 0$. Find x_k and $\lim_{k \rightarrow \infty} x_k$.

Fixed point. Solve $x = 0.8x + 5 \Rightarrow x^* = 25$.

Error form. Let $e_k := x^* - x_k$. Then $e_{k+1} = x^* - x_{k+1} = 25 - (0.8x_k + 5) = 0.8(25 - x_k) = 0.8e_k$. Hence $e_k = 0.8^k e_0 = 25 \cdot 0.8^k$, so

$$x_k = x^* - e_k = 25(1 - 0.8^k) \xrightarrow{k \rightarrow \infty} 25.$$

CDC view. With $\diamond \Phi = |x_k - x^*|$, each step has factor certificate $\Phi_{k+1} \leq 0.8 \Phi_k$.

Key Takeaways

- Geometric sums/decay give exact closed forms and explicit error bounds.
- Remainder (tail) control: $R_N = a \frac{r^{N+1}}{1-r}$ for $|r| < 1$.
- Linear recurrences $x_{k+1} = ax_k + b$ reduce to geometric decay of the error toward the fixed point.

Try It Now

1) Finite sum. Compute $\sum_{n=0}^9 5(0.9)^n$.

Answer: $5 \frac{1 - 0.9^{10}}{1 - 0.9}$.

2) Tail target. For $a = 2$, $r = \frac{3}{5}$, find N with $R_N < 10^{-4}$.

Answer: $R_N = 2 \cdot \frac{(3/5)^{N+1}}{1 - 3/5} = 5 \cdot (3/5)^{N+1} < 10^{-4} \Rightarrow (3/5)^{N+1} < 2 \cdot 10^{-5}$; take logs.

3) Recurrence. Solve $x_{k+1} = 0.6x_k + 12$, $x_0 = 30$; give $\lim x_k$.

Answer: $x^* = 12/(1 - 0.6) = 30$; $e_k = 0.6^k(30 - 30) = 0$ so $x_k \equiv 30$ (already at fixed point).

On-Page Reference

Geometric: $\sum_{n=0}^N ar^n = a \frac{1 - r^{N+1}}{1 - r}$; if $|r| < 1$, sum to ∞ is $\frac{a}{1 - r}$; tail $R_N = a \frac{r^{N+1}}{1 - r}$.

Monotone conv.: bounded increasing/decreasing sequences converge.

Linear recurrences: $x_{k+1} = ax_k + b \Rightarrow$ fixed point $x^* = \frac{b}{1 - a}$ (if $a \neq 1$), error $|x_k - x^*| = |a|^k |x_0 - x^*|$.

Alignment Notes

Assumptions: $|r| < 1$ for infinite geometric sums; consistent units in applications. **Composes with:** exponential/log (doubling/half-life), CDC factor bounds. **CDC connector:** taking $\diamond \Phi$ as error to a fixed point yields a factor certificate $\Phi_{k+1} \leq |a| \Phi_k$; the tail formula gives explicit k for a desired ε .

10. Limits by Intuition

Context and Necessity

Limits formalize “approaching” values and justify continuity, the algebra of approximations, and the special trigonometric limits that are used constantly. We keep the presentation operational: limit laws, one-sided limits, continuity at a point, removable vs. jump vs. infinite discontinuities, and the Squeeze idea.

10.1 Definitions and Laws

Definition 10.1 (Limit (informal operational form)). We write $\lim_{x \rightarrow a} f(x) = L$ to mean that the values of $f(x)$ can be made as close as desired to L by taking x sufficiently close (but not equal) to a . One-sided limits $\lim_{x \uparrow a}$ (from the left) and $\lim_{x \downarrow a}$ (from the right) are defined similarly.

Lemma 10.2 (Limit laws (where defined)). If $\lim_{x \rightarrow a} f = L$ and $\lim_{x \rightarrow a} g = M$, then

$$\lim_{x \rightarrow a} (f \pm g) = L \pm M, \quad \lim_{x \rightarrow a} (fg) = LM, \quad \lim_{x \rightarrow a} \frac{f}{g} = \frac{L}{M} \text{ if } M \neq 0.$$

Definition 10.3 (Continuity at a point). f is continuous at a if $\lim_{x \rightarrow a} f(x) = f(a)$ and both sides exist.

Lemma 10.4 (Squeeze (Sandwich) Principle). If $g(x) \leq f(x) \leq h(x)$ is near a (excluding possibly a) and $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} f(x) = L$.

10.2 Worked Example A (algebraic cancellation)

Example 10.5 (Removable discontinuity). Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.

Factor/cancel. For $x \neq 1$, $\frac{x^2 - 1}{x - 1} = \frac{(x-1)(x+1)}{x-1} = x + 1$. Thus the limit is $\boxed{2}$. (At $x = 1$ the original is undefined; define $f(1) = 2$ to make it continuous.)

10.3 Worked Example B (the standard trig limit)

Example 10.6 ($\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ (accepted fact)). Geometric or series arguments show $\frac{\sin x}{x} \rightarrow 1$ as $x \rightarrow 0$. Two consequences:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0, \quad \lim_{x \rightarrow 0} \frac{\sin(ax)}{x} = a.$$

CDC view. For small $|x|$, $\left| \frac{\sin x}{x} - 1 \right| \leq |x|/3$ (a crude bound), so choosing $|x| < \varepsilon/2$ ensures error $< \varepsilon$.

10.4 Worked Example C (piecewise and one-sided limits)

Example 10.7 (Match one-sided limits for continuity). Let

$$f(x) = \begin{cases} x^2 - 1, & x < 1, \\ mx + b, & x \geq 1. \end{cases}$$

Find m, b so that f is continuous at $x = 1$ and $\lim_{x \rightarrow 1} f'(x)$ exists (intuitive slope match).

Continuity. Left value at 1: $1^2 - 1 = 0$. Right value $m \cdot 1 + b$. Set $m + b = 0$.

Slope match (optional intuition). Left slope near 1 is $2x \rightarrow 2$. Match right slope $m = 2$, then $b = -2$. So choose $\boxed{m = 2, b = -2}$.

Key Takeaways

- Cancellation exposes the value hidden behind a removable hole.
- Limit laws let you compute systematically when individual limits exist.
- The standard trig limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ underpins small-angle approximations.
- Continuity demands agreement of left/right limits with the function value.

Try It Now

1) Algebraic limit. $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2}$.

Answer: Factor/cancel $\rightarrow x - 2$; evaluate at -2 gives -4 .

2) Trig limit. $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x}$.

Answer: 3.

3) One-sided limits. For $f(x) = \begin{cases} \sin x/x, & x \neq 0 \\ A, & x = 0 \end{cases}$, choose A for continuity.

Answer: $A = 1$.

On-Page Reference

Laws: linearity, product, quotient (limit of denominator $\neq 0$).

Squeeze: $g \leq f \leq h$, limits of g, h equal $\Rightarrow$ limit of f .

Algebraic trick: factor/cancel to remove holes before taking limits.

Special limits: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow 0} (1 + \frac{1}{x})^x = e$ (advanced).

Continuity: left limit = right limit = value at the point.

Alignment Notes

Assumptions: Standard trig limit accepted; algebraic manipulations valid away from the point. **Composes with:** rational/piecewise analysis; CDC halving on intervals for existence proofs. **CDC connector:** In practice we enforce an ε -snap: choose $|x - a| < \delta$ so the approximation error is $< \varepsilon$; for iterative methods, $\blacklozenge \Phi$ as “distance to a ” with factor steps yields explicit δ/k choices for any target ε .

Part III

Act III — Edge of Calculus

11. Rate of Change (Derivatives)

Context & Necessity

Rates convert differences into instantaneous information. The derivative formalizes “slope at a point” and supplies linear approximations used in modeling, control, and optimization. For CDC later, gradients provide certified descent under explicit step-size conditions; here we establish the definitions and the operational facts: difference quotients, the derivative as a limit, units, and continuity implications.

11.1 Definitions and Basic Facts

Definition 11.1 (Average rate of change on an interval). For a function f and x with increment $h \neq 0$,

$$\text{Average rate on } [x, x+h] = \frac{f(x+h) - f(x)}{h}.$$

Units: (units of f)/(units of x).

Definition 11.2 (Derivative at a point). If the limit

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exists (finite), f is *differentiable* at x , with derivative $f'(x)$. Notation: $\frac{dy}{dx}$ at x when $y = f(x)$, or $\left. \frac{d}{dx} f(x) \right|_x$.

Lemma 11.3 (Derivative $\Rightarrow$ continuity). If $f'(a)$ exists, then f is continuous at a .

Sketch. Write $f(a+h) - f(a) = \left(\frac{f(a+h) - f(a)}{h} \right) h$. As $h \rightarrow 0$, the bracketed term $\rightarrow f'(a)$ and $h \rightarrow 0$, so the product $\rightarrow 0$. $\square$

Lemma 11.4 (Basic cases). For constants c and linear functions $mx + b$:

$$(c)' = 0, \quad (mx + b)' = m.$$

Lemma 11.5 (Quadratic via the definition). For $f(x) = x^2$, the derivative exists for all x and $f'(x) = 2x$.

Proof.

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \xrightarrow{h \rightarrow 0} 2x.$$

$\square$

11.2 Worked Example A (from the definition: tangent slope)**Example 11.6** (Slope of x^2 at $x = 3$). Using the limit definition,

$$f'(3) = \lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h} = \lim_{h \rightarrow 0} \frac{6h + h^2}{h} = \lim_{h \rightarrow 0} (6 + h) = \boxed{6}.$$

The tangent line at $x = 3$ is $y = f(3) + f'(3)(x - 3) = 9 + 6(x - 3)$.**11.3 Worked Example B (non-differentiable cusp)****Example 11.7** ($|x|$ is not differentiable at 0). For $f(x) = |x|$, the difference quotients from left/right are

$$\frac{f(0+h) - f(0)}{h} = \frac{|h|}{h} = \begin{cases} +1, & h > 0, \\ -1, & h < 0, \end{cases}$$

so the two one-sided limits disagree; $f'(0)$ does not exist. For $x \neq 0$, $f'(x) = \text{sgn}(x)$.**11.4 Worked Example C (data & symmetric difference estimate)****Example 11.8** (Estimating a derivative from samples). Suppose a position $s(t)$ (meters) is sampled:

t (s)	1.9	2.0	2.1	2.2
s (m)	3.610	3.800	3.999	4.210

Estimate $v(2.0) = s'(2.0)$ using the symmetric difference $s'(2.0) \approx \frac{s(2.1) - s(1.9)}{0.2}$.*Calculation.* $\frac{3.999 - 3.610}{0.2} = \frac{0.389}{0.2} = 1.945$ m/s.*Units & accuracy.* Symmetric differences cancel first-order error terms and are preferable to one-sided differences when interior samples are available.**11.5 Linear Approximation (operational form)****Lemma 11.9** (Local linearization). If $f'(a)$ exists, then near a ,

$$f(a + \Delta) \approx f(a) + f'(a) \Delta.$$

The right-hand side is the tangent-line approximation; the error is $o(\Delta)$ when f' is continuous at a .

Key Takeaways

- The derivative is the limit of average rates; it implies continuity.
- Compute by definition for simple cases; use symmetric differences for numerical estimates.
- Linearization $f(a + \Delta) \approx f(a) + f'(a)\Delta$ is the default local model.

Try It Now

1) Definition practice. Use the limit definition to find $(x^2 + 3x)'$ at a general x .

Answer: Expand $(x + h)^2 + 3(x + h) - [x^2 + 3x] = 2xh + h^2 + 3h$; divide by h and take $h \rightarrow 0$ to get $2x + 3$.

2) Differentiability check. Does $f(x) = |x - 2|$ have a derivative at $x = 2$? If not, give one-sided slopes.

Answer: No. Left slope = -1 , right slope = $+1$.

3) Numerical slope. Given $g(5.0) = 11.2$, $g(5.1) = 11.55$, $g(4.9) = 10.85$, estimate $g'(5.0)$. Units same as g per unit of x .

Answer: $\frac{11.55 - 10.85}{0.2} = 3.5$ (units of g per x).

On-Page Reference

Average rate: $\frac{f(x+h) - f(x)}{h}$. **Derivative:** $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$. **Continuity:** f' exists $\Rightarrow f$ continuous.

Basic: $(c)' = 0$, $(mx + b)' = m$, $(x^2)' = 2x$ (by definition).

Symmetric difference: $f'(a) \approx \frac{f(a+h) - f(a-h)}{2h}$.

Linearization: $f(a + \Delta) \approx f(a) + f'(a)\Delta$ (tangent line).

Alignment Notes

Assumptions: Real-valued functions on real domains; limits computed where they exist.
Composes with: upcoming derivative rules (product, chain), optimization, and error bounds.
CDC connector: In gradient-based updates $x \leftarrow x - \eta \nabla f(x)$, a suitable step size η certifies a   decrease in a potential (the objective f) on each step; this yields token/factor-style bounds in later chapters.

12. Derivative Rules

Context & Necessity

The limit definition establishes what a derivative *is*; rules make it *usable*. Linearity, product/quotient, and the chain rule let us differentiate compositions quickly. We also record the elementary derivatives (power, trig, exp, log, inverse trig). These tools feed linearization, optimization, and later CDC descent steps with explicit guarantees on change per step.

12.1 Rules (statements and minimal proofs)

Lemma 12.1 (Linearity). *For differentiable f, g at x , and constants a, b :*

$$(af + bg)' = af' + bg'.$$

Lemma 12.2 (Power rule (integers)). *If $n \in \mathbb{Z}$ and $f(x) = x^n$ on its domain, then $f'(x) = nx^{n-1}$.*

Sketch. For $n = 1$ it is trivial; for $n \in \mathbb{N}$ use the binomial expansion $(x+h)^n - x^n = \binom{n}{1}x^{n-1}h + \cdots + h^n$, divide by h , and take $h \rightarrow 0$. For $n = -m$ with $m \in \mathbb{N}$, apply the quotient rule to $x^{-m} = 1/x^m$. $\square$

Lemma 12.3 (Product rule). *If f, g are differentiable at x ,*

$$(fg)' = f'g + fg'.$$

Standard trick. Add and subtract $f(x)g(x+h)$ in the numerator of the difference quotient for $f(x)g(x)$, then split into two limits. $\square$

Lemma 12.4 (Quotient rule). *If $g(x) \neq 0$ near x and f, g differentiable,*

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Proof. Apply the product rule to $f \cdot g^{-1}$ and use $(g^{-1})' = -g'/g^2$ (chain rule). $\square$

Lemma 12.5 (Chain rule). *If $y = f(u)$ is differentiable at $u = g(x)$ and $u = g(x)$ is differentiable at x , then y is differentiable at x and*

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(g(x)) g'(x).$$

12.2 Elementary derivatives

For x in each function's domain:

$$\begin{array}{ll}
 (c)' &= 0 \\
 (e^x)' &= e^x \\
 (\ln x)' &= \frac{1}{x} \quad (x > 0) \\
 (\sin x)' &= \cos x \\
 (\tan x)' &= \sec^2 x \\
 (\csc x)' &= -\csc x \cot x \\
 (\arcsin x)' &= \frac{1}{\sqrt{1-x^2}} \quad (|x| < 1) \\
 (\arctan x)' &= \frac{1}{1+x^2}
 \end{array}
 \qquad
 \begin{array}{ll}
 (x^n)' &= nx^{n-1} \quad (n \in \mathbb{Z}) \\
 (a^x)' &= (\ln a) a^x \quad (a > 0, a \neq 1) \\
 (\log_a x)' &= \frac{1}{x \ln a} \quad (x > 0) \\
 (\cos x)' &= -\sin x \\
 (\sec x)' &= \sec x \tan x \\
 (\cot x)' &= -\csc^2 x \\
 (\arccos x)' &= -\frac{1}{\sqrt{1-x^2}} \quad (|x| < 1)
 \end{array}$$

12.3 Worked Example A (product + chain)

Example 12.6 (Differentiate $y = (x^2 + 1)^5 e^{3x}$). Let $u(x) = (x^2 + 1)^5$ and $v(x) = e^{3x}$.

$$y' = u'v + uv', \quad u' = 5(x^2 + 1)^4 \cdot (2x), \quad v' = 3e^{3x}.$$

Hence

$$y' = (x^2 + 1)^4 e^{3x} (10x + 3(x^2 + 1)).$$

12.4 Worked Example B (chain on chain)

Example 12.7 (Differentiate $y = \sin(x^2 + 3x)$ and $y = \ln(\cos(2x))$). For $y = \sin(g(x))$ with $g(x) = x^2 + 3x$,

$$y' = \cos(g(x)) g'(x) = \cos(x^2 + 3x) (2x + 3).$$

For $y = \ln(\cos(2x))$, write $y = \ln(h(x))$ with $h(x) = \cos(2x)$,

$$y' = \frac{h'(x)}{h(x)} = \frac{-\sin(2x) \cdot 2}{\cos(2x)} = -2 \tan(2x),$$

(valid where $\cos(2x) \neq 0$).

12.5 Worked Example C (implicit differentiation)

Example 12.8 (Circle: $x^2 + y^2 = 25$). Differentiate both sides with respect to x :

$$2x + 2y \frac{dy}{dx} = 0 \quad \Rightarrow \quad \boxed{\frac{dy}{dx} = -\frac{x}{y}} \quad (y \neq 0).$$

At $(3, 4)$, $dy/dx = -3/4$. Tangent line: $y - 4 = (-3/4)(x - 3)$.

12.6 Linearization examples

Example 12.9 (Approximate $\sqrt{4.1}$ and $\ln(1.05)$). For $f(x) = \sqrt{x}$ at $a = 4$, $f'(x) = \frac{1}{2\sqrt{x}}$, so $f'(4) = \frac{1}{4}$. With $\Delta = 0.1$,

$$\sqrt{4.1} \approx f(4) + f'(4)\Delta = 2 + \frac{1}{4} \cdot 0.1 = 2.025.$$

For $g(x) = \ln x$ at $a = 1$, $g'(1) = 1$, so $\ln(1.05) \approx 0.05$.

Key Takeaways

- Use linearity freely; for products/quotients/compositions, select the right rule before differentiating.
- Chain rule is the workhorse: differentiate outside, multiply by the derivative of the inside.
- Keep domains explicit (e.g., $x > 0$ for $\ln x$, $\cos x \neq 0$ where $\tan x$ appears).
- Linearization provides fast, auditable approximations near a base point.

Try It Now

1) Chain. $y = \sin(x^2) \cos(3x)$. Differentiate.

Answer: $y' = 2x \cos(x^2) \cos(3x) - 3 \sin(x^2) \sin(3x)$.

2) Quotient. $y = \frac{\ln x}{x^2}$ for $x > 0$.

Answer: $y' = \frac{(1/x) \cdot x^2 - \ln x \cdot 2x}{x^4} = \frac{1 - 2x \ln x}{x^3}$.

3) Implicit. $x^3 + y^3 = 9xy$. Find dy/dx .

Answer: $3x^2 + 3y^2 y' = 9(y + xy') \Rightarrow (3y^2 - 9x)y' = 9y - 3x^2 \Rightarrow y' = \frac{9y - 3x^2}{3y^2 - 9x}$ (where denominator $\neq 0$).

4) Tangent line. $y = xe^x$ at $x = 0$.

Answer: $y' = e^x + xe^x$; at 0, $y = 0$, $y' = 1$; tangent $y = x$.

On-Page Reference

Rules: linearity; $(x^n)' = nx^{n-1}$; $(fg)' = f'g + fg'$; $(f/g)' = (f'g - fg')/g^2$; chain: $(f \circ g)' = (f' \circ g) \cdot g'$.

Elementary: $(e^x)' = e^x$, $(a^x)' = \ln(a)a^x$; $(\ln x)' = 1/x$; trig and inverse-trig derivatives as listed above.

Implicit: differentiate both sides; collect y' ; solve; state domain conditions.

Linearization: $f(a + \Delta) \approx f(a) + f'(a)\Delta$ near a .

Alignment Notes

Assumptions: Each rule requires differentiability where used; quotient demands $g \neq 0$.

Composes with: optimization (critical points), error bounds, and numerical methods.

CDC connector: With objective $F(x)$ as $\diamond \Phi$, a step $x \leftarrow x - \eta F'(x)$ admits a certified decrease for sufficiently small η (factor-style for Lipschitz gradients, token-style for bounded-away-from-zero slopes on intervals), producing explicit step bounds later.

13. Accumulation (Integrals)

Context & Necessity

Integrals quantify accumulated change: area under a rate, mass from density, work from force. Operationally we use two views:

- (i) *Riemann sums* (add many small contributions),
- (ii) *anti-derivatives* via the Fundamental Theorem of Calculus (FTC).

Both connect to CDC: partition size or step size becomes a nonnegative potential $\blacklozenge \Phi$ refining the mesh halves $\blacklozenge \Phi$ and shrinks the error by a factor.

13.1 Definitions and Core Laws

Definition 13.1 (Definite integral (Riemann)). For f bounded on $[a, b]$, partition $a = x_0 < \cdots < x_n = b$, pick $t_k \in [x_{k-1}, x_k]$, and set $S = \sum_{k=1}^n f(t_k) \Delta x_k$ with $\Delta x_k = x_k - x_{k-1}$. If all such sums approach the same limit as $\max_k \Delta x_k \rightarrow 0$, we write

$$\int_a^b f(x) dx = \lim_{\max \Delta x_k \rightarrow 0} \sum_{k=1}^n f(t_k) \Delta x_k.$$

Definition 13.2 (Indefinite integral / anti-derivative). A function F is an anti-derivative of f on an interval if $F' = f$. We write $\int f(x) dx = F(x) + C$ for an arbitrary constant C .

Theorem 13.3 (Fundamental Theorem of Calculus (FTC)). *Let f be continuous on $[a, b]$.*

(a) (FTC1) $F(x) := \int_a^x f(t) dt$ is differentiable and $F'(x) = f(x)$.

(b) (FTC2) If F is any antiderivative of f , then $\int_a^b f(x) dx = F(b) - F(a)$.

Lemma 13.4 (Linearity and additivity). *For integrable f, g and scalars α, β :*

$$\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g, \quad \int_a^c f = \int_a^b f + \int_b^c f.$$

13.2 Elementary Anti-derivatives

For x in each function's domain:

$$\begin{array}{ll}
 \int 0 \, dx &= C \\
 \int x^n \, dx &= \frac{x^{n+1}}{n+1} + C \quad (n \neq -1) \\
 \int e^x \, dx &= e^x + C \\
 \int \sin x \, dx &= -\cos x + C \\
 \int \sec^2 x \, dx &= \tan x + C \\
 \int 1 \, dx &= x + C \\
 \int \frac{1}{x} \, dx &= \ln |x| + C \\
 \int a^x \, dx &= \frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1) \\
 \int \cos x \, dx &= \sin x + C \\
 \int \frac{1}{1+x^2} \, dx &= \arctan x + C
 \end{array}$$

13.3 Worked Example A (Riemann sum $\rightarrow$ number $\rightarrow$ exact check)

Example 13.5 (Midpoint rule on $[0, 2]$). Approximate $I = \int_0^2 (x^2 + 1) \, dx$ with $n = 4$ equal sub-intervals (midpoints), then compare to the exact value.

Midpoints. $\Delta x = \frac{2-0}{4} = 0.5$; midpoints $m_k = 0.25, 0.75, 1.25, 1.75$.

$$S_M = \sum_{k=1}^4 (m_k^2 + 1) \Delta x = (0.25^2 + 1 + 0.75^2 + 1 + 1.25^2 + 1 + 1.75^2 + 1) \cdot 0.5 = \boxed{3.75}.$$

Exact (FTC2). $\int (x^2 + 1) \, dx = \frac{x^3}{3} + x$, so $I = \left[\frac{x^3}{3} + x \right]_0^2 = \frac{8}{3} + 2 = \frac{14}{3} \approx 4.666 \dots$

Comment. Midpoint underestimates here; refining to $n \mapsto 2n$ halves Δx and (for smooth f) reduces midpoint error by about a factor of 4.

13.4 Worked Example B (net change from a rate)

Example 13.6 (Distance from velocity). A car's velocity (m/s) is $v(t) = 3t^2 - 2t + 1$ on $t \in [0, 4]$ (seconds). Find the net displacement.

Integrate. $s(4) - s(0) = \int_0^4 v(t) \, dt = [t^3 - t^2 + t]_0^4 = (64 - 16 + 4) - 0 = \boxed{52 \text{ m}}.$

Units. (m/s)·s = meters.

13.5 Worked Example C (area between curves)

Example 13.7 (Area between $\sin x$ and $\cos x$ on $[0, \pi/2]$). The area is $A = \int_0^{\pi/2} |\cos x - \sin x| \, dx$. On $[0, \pi/4]$, $\cos x \geq \sin x$; on $[\pi/4, \pi/2]$, $\sin x \geq \cos x$. By symmetry,

$$A = 2 \int_0^{\pi/4} (\cos x - \sin x) \, dx = 2 [\sin x + \cos x]_0^{\pi/4} = 2 \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - (0 + 1) \right) = \boxed{2\sqrt{2} - 2}.$$

Key Takeaways

- Riemann sums formalize accumulation; FTC turns accumulation into evaluation of antiderivatives.
- Linearity and additivity let you compute piecewise or combine rates.
- Areas between curves use “upper – lower” with correct breakpoints (where the curves cross).

Try It Now

1) Exact integral. $\int_1^3 (2x - 1) dx$.

Answer: $[x^2 - x]_1^3 = (9 - 3) - (1 - 1) = 6$.

2) Midpoint estimate. Approximate $\int_0^1 e^x dx$ with $n = 2$ midpoint subintervals. Compare with exact $e - 1$.

Answer: $\Delta x = 0.5$, midpoints 0.25, 0.75: $S_M = 0.5(e^{0.25} + e^{0.75}) \approx 0.5(1.2840 + 2.1170) \approx 1.7005$; exact $e - 1 \approx 1.7183$.

3) Area between curves. Area enclosed by $y = x$ and $y = x^2$ on $[0, 1]$.

Answer: $\int_0^1 (x - x^2) dx = [\frac{x^2}{2} - \frac{x^3}{3}]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.

On-Page Reference

FTC: $\frac{d}{dx}(\int_a^x f) = f(x)$; $\int_a^b f = F(b) - F(a)$ if $F' = f$.

Linearity/additivity: pull out constants; split intervals.

Midpoint rule: $S_M = \sum f(m_k) \Delta x$; refined partitions $\Rightarrow$ smaller error.

Areas: $\int$ (upper – lower) with breakpoints at intersections.

Units: (units of rate) $\times$ (units of input) = (units of total).

Alignment Notes

Assumptions: Continuity where FTC is applied; piecewise handling at crossings.

Composes with: derivative rules (check by differentiating an anti-derivative), piecewise/rational analysis (breakpoints), and error control via mesh refinement. **CDC connector:**

Let $\blacklozenge \Phi$ be mesh size Δ ; halving the mesh ($\rightarrow \mathbf{U}$) yields a factor certificate $\checkmark \mathbf{C}$ on many quadrature errors. Stopping when the estimated error $< \varepsilon$ is the halt rule $\square \mathbf{H}$.

14. Optimization and Modeling

Context & Necessity

Optimization turns calculus into decisions: when does a quantity achieve its best (max/min) value, and where? The workflow is standard:

- (i) write the objective f on a precise domain,
- (ii) locate *critical points* (interior candidates),
- (iii) check *boundaries/constraints*, and
- (iv) certify optimality (first or second derivative tests).

In CDC terms, an objective can serve as a potential $\blacklozenge \Phi$; a step rule that reduces f and yields a one-line certificate of progress.

14.1 Definitions and Tests

Definition 14.1 (Critical point). If f is defined on an open interval and $f'(c) = 0$ or f' is undefined at c (but f is defined), then c is a *critical point*. Interior extrema occur only at critical points.

Lemma 14.2 (First Derivative Test). *Let c be a critical point of f and assume f is continuous near c .*

- *If f' changes $+$ to $-$ at c , then f has a local maximum at c .*
- *If f' changes $-$ to $+$ at c , then f has a local minimum at c .*
- *If f' does not change sign, c is not a local extremum.*

Lemma 14.3 (Second Derivative Test). *If $f'(c) = 0$ and f'' is continuous near c , then $f''(c) > 0 \Rightarrow$ local minimum; $f''(c) < 0 \Rightarrow$ local maximum. If $f''(c) = 0$, the test is inconclusive (use the First Derivative Test or higher derivatives).*

Lemma 14.4 (Boundary check is mandatory). *For closed intervals $[a, b]$, absolute extrema occur among critical points in (a, b) and endpoints a, b .*

14.2 Worked Example A (geometry under constraint)

Example 14.5 (Maximize rectangle area with fixed perimeter). A rectangle has perimeter $P > 0$. Find the dimensions with maximal area.

Model. Let sides be x and y . Constraint $2(x + y) = P \Rightarrow y = \frac{P}{2} - x$. Area $A(x) = xy = x(\frac{P}{2} - x) = \frac{P}{2}x - x^2$ on $x \in (0, \frac{P}{2})$.

Differentiate. $A'(x) = \frac{P}{2} - 2x$, so $A'(x) = 0 \Rightarrow x = \frac{P}{4}$. Then $y = \frac{P}{4}$.

Classify. $A''(x) = -2 < 0$ so this is a (strict) local, and hence global maximum.

Conclusion. The square is optimal: $x = y = \frac{P}{4}$.

14.3 Worked Example B (open-top box from a sheet)

Example 14.6 (Maximize volume by cutting corner squares). From a 20×30 sheet (units arbitrary), cut equal squares of side s from each corner and fold to make an open-top box. Maximize volume.

Model. Base $(20 - 2s) \times (30 - 2s)$, height s :

$$V(s) = s(20 - 2s)(30 - 2s) = 600s - 100s^2 + 4s^3, \quad s \in (0, 10).$$

Differentiate. $V'(s) = 600 - 200s + 12s^2$. Solve $V'(s) = 0$: $12s^2 - 200s + 600 = 0 \Rightarrow 3s^2 - 50s + 150 = 0$.

Critical points. $s = \frac{50 \pm \sqrt{700}}{6}$. Only $s \approx \frac{50 - 10\sqrt{7}}{6} \approx 3.924$ lies in $(0, 10)$.

Classify. $V''(s) = 24s - 200$; at $s \approx 3.924$, $V'' < 0 \Rightarrow$ local max. Endpoints $s \rightarrow 0^+$ or 10^- give $V \rightarrow 0$.

Conclusion. $s \approx 3.924$ maximizes V ; dimensions $(20 - 2s) \times (30 - 2s) \times s \approx 12.15 \times 22.15 \times 3.92$.

14.4 Worked Example C (closest point / constrained distance)

Example 14.7 (Closest point on $y = \sqrt{x}$ to $(3, 0)$). Minimize distance squared $D(x) = (x - 3)^2 + (\sqrt{x} - 0)^2$ for $x \geq 0$. Then $D(x) = x^2 - 5x + 9$.

Differentiate. $D'(x) = 2x - 5$. Set $D'(x) = 0 \Rightarrow x = \frac{5}{2}$. Since $x \geq 0$ and $D''(x) = 2 > 0$, this is a (strict) minimum.

Conclusion. Closest point $(\frac{5}{2}, \sqrt{\frac{5}{2}})$; minimum distance $= \sqrt{D(5/2)} = \sqrt{2.75} \approx 1.658$.

14.5 Operational Patterns (for modeling problems)

- **Reduce to one variable.** Use constraints to eliminate variables early.
- **Domain is part of the problem.** Include physical bounds (e.g., $s \in (0, 10)$).
- **Classify candidates.** First/second derivative tests; always check boundaries.
- **Units and scaling.** Keep units consistent; consider nondimensionalizing where helpful.

Key Takeaways

- Interior extrema occur only at critical points; absolute extrema also consider boundaries.
- First Derivative Test uses sign changes; Second Derivative Test uses concavity.
- Constrained models typically become single-variable problems via substitution.

Try It Now

1) Parabola rectangle. Maximize the area of a rectangle symmetric about the y -axis under $y = 6 - x^2$ with top corners on the curve.

Answer: Half-width $a > 0$, height $h = 6 - a^2$, area $A = 2a(6 - a^2)$. $A'(a) = 12 - 6a^2 = 0 \Rightarrow a = \sqrt{2}$; $A_{\max} = 2\sqrt{2}(6 - 2) = 8\sqrt{2}$.

2) Revenue model. Demand $q(p) = 100 - 5p$ (units sold at price p). Maximize revenue $R(p) = pq(p)$ for $p \geq 0$.

Answer: $R(p) = 100p - 5p^2$; $R'(p) = 100 - 10p = 0 \Rightarrow p = 10$; $R'' = -10 < 0$ max.

3) Endpoint awareness. Minimize $f(x) = x + \frac{9}{x}$ on $x \in [1, 5]$.

Answer: $f'(x) = 1 - \frac{9}{x^2} = 0 \Rightarrow x = 3$. Check $f(1) = 10$, $f(3) = 6$, $f(5) = 5 + \frac{9}{5} = 6.8$. Minimum at $x = 3$.

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Workflow: objective f + domain $\rightarrow$ critical points ($f' = 0$ or f' undefined) $\rightarrow$ classify (1st/2nd derivatives) $\rightarrow$ include endpoints.

1st Deriv. Test: sign change of f' across c . **2nd Deriv. Test:** $f''(c) \gtrless 0 \Rightarrow$ local min/max.

Constrained reduction: eliminate variables via the constraint; keep domain explicit.

Quadratic forms: concave down ($a < 0$) $\Rightarrow$ unique max; up ($a > 0$) $\Rightarrow$ unique min.

Alignment Notes

Assumptions: Differentiability where tests are applied; domains closed where absolute extrema are claimed. **Composes with:** derivative rules, integrals (work/energy), and modeling chapters. **CDC connector:** Using f as $\blacklozenge \Phi$, a descent update (e.g., $x \leftarrow x - \eta f'(x)$ on a 1D smooth f) admits a $\checkmark \text{C}$ certificate of decrease for appropriate η . Mixed token/factor style bounds appear once Lipschitz or strong-convexity conditions are introduced.

Part IV

Act IV — Calculus Core

15. First Differential Equations

Context & Necessity

Differential equations describe how a quantity changes in response to its current state or time: growth/decay, cooling, forcing, feedback. We focus on solvable, ground-level forms, *separable* and *linear first order*; and on how to read qualitative behavior (equilibria, stability). CDC appears via Lyapunov-style potentials: e.g., distance to equilibrium decays exponentially, yielding factor certificates and explicit time-to- ε bounds.

15.1 Definitions and Basic Forms

Definition 15.1 (Differential equation (DE), order, IVP). A (scalar) differential equation relates y and its derivatives: $F(x, y, y', \dots) = 0$. The *order* is the highest derivative present. An *initial value problem* (IVP) specifies $y(x_0) = y_0$ (and possibly more conditions for higher orders).

Definition 15.2 (Equilibrium and stability (first order)). For $y' = f(y)$ (autonomous), $y = y^*$ is an *equilibrium* if $f(y^*) = 0$. If nearby solutions move toward y^* as t increases, it is (locally) *stable*; if they move away, *unstable*.

Definition 15.3 (Separable). A first-order DE is *separable* if it can be written $y' = g(x)h(y)$, so that

$$\frac{dy}{h(y)} = g(x) dx \quad \text{and integrating gives an implicit/explicit solution.}$$

Definition 15.4 (Linear first order). A first-order DE is *linear* if it has the form

$$y' + p(x)y = q(x).$$

Its integrating factor is $\mu(x) = e^{\int p(x) dx}$, giving $(\mu y)' = \mu q$.

15.2 Worked Example A (exponential growth/decay)

Example 15.5 ($y' = ky$; doubling/half-life and CDC factor). Solve the IVP $y' = ky$, $y(0) = y_0$.

Separate. $\frac{dy}{y} = k dx \Rightarrow \ln |y| = kx + C \Rightarrow y(x) = y_0 e^{kx}$.

Characteristic times. If $k > 0$, doubling time $T_{\text{double}} = \frac{\ln 2}{k}$. If $k < 0$, half-life $T_{1/2} = \frac{\ln 2}{|k|}$.

CDC view. Take $\Phi = |y|$ (for decay $k < 0$) or deviation from equilibrium $\Phi = |y - y^*|$ with $y^* = 0$. Then $\Phi(x) = e^{kx}\Phi(0)$ gives a factor certificate:

$$\Phi(x + \Delta) \leq e^{k\Delta} \Phi(x) \quad (\text{with } e^{k\Delta} < 1 \text{ if } k < 0).$$

To reach $\Phi \leq \varepsilon$: $x \geq \frac{1}{|k|} \ln(\Phi(0)/\varepsilon)$.

15.3 Worked Example B (Newton cooling: stable equilibrium)

Example 15.6 ($y' = -k(y - T_{\text{env}})$ with $k > 0$). Let $y(t)$ be the object temperature; T_{env} is constant ambient temperature. Solve $y' = -k(y - T_{\text{env}})$, $y(0) = y_0$.

Separate. $\frac{dy}{y - T_{\text{env}}} = -k dt$. Integrate: $\ln|y - T_{\text{env}}| = -kt + C$. Thus

$$y(t) = T_{\text{env}} + (y_0 - T_{\text{env}})e^{-kt}.$$

Stability. Equilibrium $y^* = T_{\text{env}}$ is stable since $|y(t) - y^*| = |y_0 - y^*|e^{-kt} \rightarrow 0$.

CDC view. $\Phi = |y - T_{\text{env}}|$ admits factor certificates: $\Phi(t + \Delta) \leq e^{-k\Delta}\Phi(t)$. Choose Δ so $e^{-k\Delta} \leq c < 1$ to get a per-step factor bound; time-to- ε follows from logs.

15.4 Worked Example C (linear first order via integrating factor)

Example 15.7 ($y' + \frac{2}{x}y = 3x^2$, $x > 0$). Solve on $x > 0$ the linear DE with $y(1) = 0$.

Integrating factor. $p(x) = \frac{2}{x} \Rightarrow \mu(x) = e^{\int (2/x) dx} = e^{2 \ln x} = x^2$.

Multiply. $(x^2 y)' = x^2 \cdot 3x^2 = 3x^4$. Integrate: $x^2 y = \int 3x^4 dx = \frac{3}{5}x^5 + C$.

Solve. $y(x) = \frac{3}{5}x^3 + \frac{C}{x^2}$. Use $y(1) = 0 \Rightarrow 0 = \frac{3}{5} + C \Rightarrow C = -\frac{3}{5}$. Hence

$$y(x) = \frac{3}{5}\left(x^3 - \frac{1}{x^2}\right), \quad x > 0.$$

Check. Substitute back and verify $y' + \frac{2}{x}y = 3x^2$.

15.5 Qualitative Tools (quick)

- **Slope field (direction field):** draw small line elements with slope $y' = f(x, y)$ to visualize trajectories.
- **Nullclines and equilibria (autonomous $y' = f(y)$):** roots of $f(y)$ are equilibria; sign of f between them shows drift.
- **Stability test (autonomous):** if $f'(y^*) < 0$, then y^* is (locally) stable; if $f'(y^*) > 0$, unstable.

Key Takeaways

- Separable: move all y -terms left, x -terms right; integrate both sides.
- Linear first order: $y' + p(x)y = q(x)$ with integrating factor $\mu = e^{\int p}$ so $(\mu y)' = \mu q$.
- Equilibria and stability give qualitative predictions; exponential convergence supplies explicit factor bounds.

Try It Now

1) Separable. Solve $y' = xy$, $y(0) = 2$.

Answer: $\frac{dy}{y} = x dx \Rightarrow \ln |y| = \frac{x^2}{2} + C \Rightarrow y = 2e^{x^2/2}$.

2) Logistic (autonomous). $y' = ry(1 - y/K)$, $y(0) = y_0 \in (0, K)$. Write the explicit solution.

Answer (form): $y(t) = \frac{K}{1 + Ae^{-rt}}$, $A = \frac{K - y_0}{y_0}$.

3) Linear with constant coefficients. Solve $y' - 2y = e^x$.

Answer: $\mu = e^{-2x} \Rightarrow (ye^{-2x})' = e^{-x} \Rightarrow ye^{-2x} = -e^{-x} + C \Rightarrow y = -e^x + Ce^{2x}$.

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Separable: $y' = g(x)h(y) \Rightarrow \int \frac{dy}{h(y)} = \int g(x) dx + C$.

Linear 1st order: $y' + p(x)y = q(x)$, $\mu = e^{\int p}$, $(\mu y)' = \mu q \Rightarrow y = \mu^{-1}(\int \mu q + C)$.

Exponential times: $y' = ky \Rightarrow y = y_0 e^{kx}$; $T_{\text{double}} = \ln 2/k$, $T_{1/2} = \ln 2/|k|$.

Stability (autonomous): $f'(y^*) < 0 \Rightarrow \text{stable}$; $f'(y^*) > 0 \Rightarrow \text{unstable}$.

CDC link: choose $\Phi = |y - y^*|$ to get $\Phi(t) = e^{-kt}\Phi(0)$ (factor bound).

Alignment Notes

Assumptions: Domains where integrals/antiderivatives are valid; $x > 0$ where $\ln x$ appears (e.g., $\mu = e^{\int (2/x)dx}$).

Composes with: optimization (Lyapunov functions), numerical methods (Euler/Runge-Kutta), and later CDC chapters on certified step sizes.

CDC connector: With $\diamond \Phi = |y - y^*|$, exact solutions give factor-style decay. Discrete solvers gain token/factor certificates by picking step sizes that guarantee Φ decreases per step.

16. Limits and Continuity (Precise)

Context & Necessity

The ε - δ definition turns “approaches” into mathematics. It justifies the limit laws, continuity claims, and theorems (IVT, EVT) we use implicitly earlier. For CDC, precise limits certify that a small *input control* ($|x - a| < \delta$) implies a small *output effect* ($|f(x) - L| < \varepsilon$); paired with interval halving, this yields auditable step counts to meet tolerances.

16.1 Definitions

Definition 16.1 (Limit ε - δ). For a function f defined on a punctured neighborhood of a ,

$$\lim_{x \rightarrow a} f(x) = L$$

means: $\forall \varepsilon > 0 \exists \delta > 0 \forall x (0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon)$.

Definition 16.2 (One-sided limits). $\lim_{x \uparrow a} f(x) = L$ requires $0 < a - x < \delta \Rightarrow |f(x) - L| < \varepsilon$. Similarly for $\lim_{x \downarrow a}$.

Definition 16.3 (Continuity at a point). f is continuous at a if it is defined at a and $\lim_{x \rightarrow a} f(x) = f(a)$.

16.2 Characterizations and Laws

Lemma 16.4 (Sequential characterization). $\lim_{x \rightarrow a} f(x) = L$ iff for every sequence (x_n) with $x_n \neq a$ and $x_n \rightarrow a$, we have $f(x_n) \rightarrow L$.

Lemma 16.5 (Limit laws). If $\lim_{x \rightarrow a} f = L$ and $\lim_{x \rightarrow a} g = M$, then

$$\lim_{x \rightarrow a} (f \pm g) = L \pm M, \quad \lim_{x \rightarrow a} (fg) = LM, \quad \lim_{x \rightarrow a} \frac{f}{g} = \frac{L}{M} \quad (M \neq 0).$$

Lemma 16.6 (Composite limits). If $\lim_{x \rightarrow a} g(x) = b$ and f is continuous at b , then $\lim_{x \rightarrow a} f(g(x)) = f(b)$.

Lemma 16.7 (Squeeze ”Sandwich” Principle). If $g(x) \leq f(x) \leq h(x)$ near a and $\lim_{x \rightarrow a} g = \lim_{x \rightarrow a} h = L$, then $\lim_{x \rightarrow a} f = L$.

16.3 Continuity Theorems on Closed Intervals

Theorem 16.8 (Intermediate Value Theorem (IVT)). *If f is continuous on $[a, b]$ and y lies between $f(a)$ and $f(b)$, then $\exists c \in [a, b]$ with $f(c) = y$.*

Theorem 16.9 (Extreme Value Theorem (EVT)). *If f is continuous on $[a, b]$, then f attains a maximum and a minimum on $[a, b]$.*

Theorem 16.10 (Heine–Cantor (uniform continuity on compact sets)). *If f is continuous on $[a, b]$, then f is uniformly continuous there: $\forall \varepsilon > 0 \exists \delta > 0$ such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$ for all $x, y \in [a, b]$ (single δ works for the whole interval).*

16.4 Worked Example A (constructing a δ)

Example 16.11 ($\lim_{x \rightarrow a} (mx + b) = ma + b$ with δ in hand). Fix $\varepsilon > 0$. For $f(x) = mx + b$,

$$|f(x) - (ma + b)| = |m||x - a|.$$

Choose $\delta = \varepsilon/|m|$ for $m \neq 0$ (if $m = 0$, any δ works). Then $0 < |x - a| < \delta \Rightarrow |f(x) - (ma + b)| < \varepsilon$.

16.5 Worked Example B (continuity of polynomials and rationals)

Example 16.12 (Polynomials are continuous; rationals where defined). By limit laws and the fact $x \mapsto x$ is continuous, sums/products preserve continuity. Thus any polynomial p is continuous everywhere; any rational $r = p/q$ is continuous where $q \neq 0$. E.g., $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2$.

16.6 Worked Example C (IVT + CDC step count: bisection)

Example 16.13 (Existence and certified search for a root). Assume f continuous on $[L_0, R_0]$ with $f(L_0)f(R_0) < 0$. By IVT, there exists x^* with $f(x^*) = 0$. Define

◆ $\Phi = R - L$, **G**: $f(L)f(R) \leq 0$, $\rightarrow$ **U**: bisect and keep the sign-change half, □ **H**: $\Phi \leq \varepsilon$.

Each step satisfies $\Phi' \leq \frac{1}{2}\Phi$ (factor certificate). After k steps, $\Phi_k \leq \Phi_0/2^k$, so to obtain interval width $\leq \varepsilon$ it suffices that

$$k \geq \left\lceil \log_2 \frac{\Phi_0}{\varepsilon} \right\rceil.$$

This pairs a *qualitative* existence proof (IVT) with a *quantitative* CDC bound.

16.7 Types of Discontinuities (taxonomy)

- **Removable**: $\lim_{x \rightarrow a} f(x)$ exists, but $f(a)$ is missing or unequal.
- **Jump**: one-sided limits exist and differ.
- **Infinite**: $|f(x)| \rightarrow \infty$ as $x \rightarrow a$ (vertical asymptote).

Key Takeaways

- ε - δ definitions make limit claims checkable and move us beyond heuristics.
- Continuity is a limit equality at the point; IVT/EVT require continuity on closed intervals.
- Pair IVT with bisection to get existence *and* explicit step counts toward any tolerance.

Try It Now

1) Delta design. Prove $\lim_{x \rightarrow 2} (3x - 5) = 1$ by giving a $\delta(\varepsilon)$.

Answer: $|3x - 5 - 1| = 3|x - 2|$. Take $\delta = \varepsilon/3$.

2) Continuity domain. Where is $f(x) = \frac{x^2 - 4}{x^2 - 1}$ continuous? Compute $\lim_{x \rightarrow 1} f(x)$.

Answer: Continuous on $\mathbb{R} \setminus \{\pm 1\}$. Limit does not exist at 1 (blows up).

3) IVT application. Show $g(x) = x^3 - 7x + 1$ has a real root in $[0, 1]$ and bound bisection steps to reach width $\leq 10^{-4}$.

Answer: $g(0) = 1 > 0$, $g(1) = -5 < 0 \Rightarrow$ root in $[0, 1]$. Need $k \geq \lceil \log_2(1/10^{-4}) \rceil = \lceil \log_2(10^4) \rceil \approx \lceil 13.29 \rceil = 14$.

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Limit: $\forall \varepsilon \exists \delta : 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

Continuity: $\lim_{x \rightarrow a} f(x) = f(a)$. **Laws:** linearity, product, quotient (denominator limit $\neq 0$).

Sequential form: $x_n \rightarrow a$, $x_n \neq a \Rightarrow f(x_n) \rightarrow L$.

IVT/EVT: continuous on $[a, b] \Rightarrow$ intermediate values attained; extrema attained.

Bisection steps: $k \geq \lceil \log_2(\text{width}_0/\varepsilon) \rceil$.

Alignment Notes

Assumptions: Standard real analysis setting; continuity on closed intervals for IVT/EVT.

Composes with: earlier algebraic/trig limits, rational/piecewise analysis, and Act III methods. **CDC connector:** Tolerance ε is a target on $\diamond \Phi$; halving provides factor

certificates and an explicit audit trail from hypothesis (continuity + sign change) to step count.

17. Differentiation — Further Topics

Context & Necessity

The limit definition and basic rules make derivatives computable; the theorems here make them *powerful*. Rolle's Theorem and the Mean Value Theorem (MVT) turn local slopes into global statements (monotonicity, Lipschitz bounds). Second-derivative tests certify concavity/inflection and sharpen optimization. L'Hospital's Rule converts indeterminate forms into tractable limits. For CDC later, these provide crisp conditions under which a chosen potential $\blacklozenge \Phi$ is guaranteed to drop by a certified amount.

17.1 Rolle, MVT, and Consequences

Theorem 17.1 (Rolle's Theorem). *If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.*

Theorem 17.2 (Mean Value Theorem (MVT)). *If f is continuous on $[a, b]$ and differentiable on (a, b) , then*

$$\exists c \in (a, b) \text{ such that } f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Lemma 17.3 (Monotonicity from the derivative). *If f' exists on (a, b) and $f'(x) \geq 0$ for all x there, then f is non-decreasing on $[a, b]$. If $f'(x) \leq 0$, then f is non-increasing.*

Sketch using MVT. For $x < y$, Theorem 17.2 gives $f(y) - f(x) = f'(c)(y - x)$ for some $c \in (x, y)$. If $f'(c) \geq 0$, then $f(y) \geq f(x)$. $\square$

Lemma 17.4 (Lipschitz bound from a derivative cap). *If f' exists on (a, b) and $|f'(x)| \leq M$ for all x in (a, b) , then*

$$|f(y) - f(x)| \leq M |y - x| \quad \text{for all } x, y \in [a, b].$$

17.2 Concavity, Inflection, and Tests

Definition 17.5 (Concavity). A function f is *concave up* on an interval if its graph lies above every tangent line there; *concave down* if it lies below. An *inflection point* is a point where concavity changes.

Lemma 17.6 (Second-derivative tests). *If f'' exists on an interval,*

- $f'' > 0 \Rightarrow f$ is concave up; $f'' < 0 \Rightarrow$ concave down.
- If $f'(c) = 0$ and $f''(c) > 0$ (resp. < 0), then f has a local min (resp. max) at c .

17.3 L'Hospital's Rule (indeterminate forms)

Theorem 17.7 (L'Hospital (0/0 or ∞/∞)). *Let f, g be differentiable on an open interval containing a (possibly excluding a). Assume $g'(x) \neq 0$ near a . If either*

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0 \quad \text{or} \quad \lim_{x \rightarrow a} |g(x)| = \infty \quad \text{and} \quad \lim_{x \rightarrow a} |f(x)| = \infty,$$

and if $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L$ exists (finite or $\pm\infty$), then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L$.

(Variants hold for one-sided limits and for $a = \pm\infty$.)

17.4 Worked Example A (MVT $\Rightarrow$ uniform bound)

Example 17.8 (Bounding change on a bounded-derivative interval). On $[0, 1]$, let $f(x) = e^x$. Since $f'(x) = e^x \leq e$ for $x \in [0, 1]$, the derivative is bounded by $M = e$. Thus for any $x, y \in [0, 1]$,

$$|e^y - e^x| \leq e |y - x|.$$

Use. This converts input tolerance into output error control; with $\blacklozenge \Phi = |y - x|$, each halving of the input span halves the output bound.

17.5 Worked Example B (sign chart via f' and f'')

Example 17.9 (Classification for $f(x) = x^3 - 3x^2 - 9x + 2$). Compute $f'(x) = 3x^2 - 6x - 9 = 3(x - 3)(x + 1)$ and $f''(x) = 6x - 6 = 6(x - 1)$.

Critical points. $f' = 0$ at $x = -1$ and $x = 3$.

- On $(-\infty, -1)$, $f' > 0$ (increasing); $(-1, 3)$, $f' < 0$ (decreasing); $(3, \infty)$, $f' > 0$ (increasing).
- Thus local max at $x = -1$, local min at $x = 3$.

Concavity/inflection. $f'' = 0$ at $x = 1$; $f'' < 0$ for $x < 1$ (concave down), $f'' > 0$ for $x > 1$ (concave up). Hence an inflection at $x = 1$ with $f(1) = 1 - 3 - 9 + 2 = -9$.

Summary. Increase $\uparrow$ on $(-\infty, -1)$ and $(3, \infty)$; decrease $\downarrow$ on $(-1, 3)$; concave $\downarrow$ for $x < 1$, concave $\uparrow$ for $x > 1$.

17.6 Worked Example C (L'Hospital in action)

Example 17.10 (Three classic limits). (a) $\lim_{x \rightarrow 0} \frac{\sin x}{x}$: $f = \sin x$, $g = x$. Then $f'/g' = \cos x/1 \rightarrow 1 \Rightarrow \text{limit} = 1$.

(b) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$: $f = 1 - \cos x$, $g = x^2$. Then $f'/g' = \sin x/(2x) \rightarrow 0/0$. Apply again: $f''/g'' = \cos x/2 \rightarrow 1/2 \Rightarrow \text{limit} = 1/2$.

(c) $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$: $f = \ln x$, $g = x$. Then $f'/g' = (1/x)/1 \rightarrow 0 \Rightarrow \text{limit} = 0$.

17.7 Operational Patterns (audit-friendly)

- **Global change bound.** If $|f'| \leq M$ on an interval, then a width- Δ bracket ensures $|f(y) - f(x)| \leq M\Delta$.
- **Monotonicity from sign.** Use the sign of f' to partition increase/decrease intervals; classify extrema with the sign change or f'' .
- **Concavity/linearization.** With f'' known, tangent-line errors are controlled (to first order) by the size of f'' on the interval.
- **Indeterminate forms.** Check hypotheses before using L'Hospital: correct form ($0/0$ or ∞/∞), differentiability, and nonzero g' near the point.

Key Takeaways

- MVT bridges local slope and global change; a bound on $|f'|$ yields Lipschitz control.
- Signs of f'/f'' organize increase/decrease and concavity/inflection, streamlining optimization.
- L'Hôpital reduces many hard limits to simpler derivative ratios when its hypotheses truly hold.

Try It Now

1) Lipschitz from derivative cap. On $[0, \pi]$, show $|\sin y - \sin x| \leq |y - x|$.

Answer: $|\sin y - \sin x| = |\cos(\xi)| |y - x| \leq 1 \cdot |y - x|$ by MVT for some ξ between x and y .

2) Classification. For $g(x) = x^4 - 4x^2$, find intervals of increase/decrease, local extrema, and inflection points.

Answer (sketch): $g' = 4x^3 - 8x = 4x(x^2 - 2)$; critical at $x \in \{0, \pm\sqrt{2}\}$. $g'' = 12x^2 - 8$; inflections at $x = \pm\sqrt{2/3}$. Use sign charts to classify.

3) L'Hospital check. Evaluate $\lim_{x \rightarrow 0} \frac{\ln(1+x) - x}{x^2}$.

Answer: $0/0$. First pass: $\frac{(1/(1+x)) - 1}{2x} = \frac{\frac{-x}{1+x}}{2x} = \frac{-1}{2(1+x)} \rightarrow -\frac{1}{2}$.

4) MVT application. If $|f'| \leq 0.2$ on $[2, 2.5]$, bound $|f(2.5) - f(2)|$.

Answer: $\leq 0.2 \cdot 0.5 = 0.1$.

On-Page Reference

Rolle/MVT: equal endpoints $\Rightarrow$ some $f'(c) = 0$; in general $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Monotonicity: sign of f' ; **Lipschitz:** $|f'| \leq M \Rightarrow |f(y) - f(x)| \leq M|y - x|$.

Concavity: $f'' \geq 0$; **2nd-deriv. test:** $f''(c) \geq 0 \Rightarrow \min/\max$ at c (if $f'(c) = 0$).

L'Hospital: only for $0/0$ or ∞/∞ with differentiability and $g' \neq 0$ near the point.

Audit tip: always state domain and hypotheses before using the theorem.

Alignment Notes

Assumptions: Continuity/differentiability where required; correct indeterminate form for L'Hospital. **Composes with:** optimization chapter, limits (precise), and integration error controls. **CDC connector:** With $\blacklozenge \Phi$ as interval width or residual size, MVT gives explicit change bounds per step, and concavity/convexity support certified descent (unique minima under strong convexity in later pages).

18. Integration Techniques

Context & Necessity

Anti-derivatives turn rates into totals (FTC). In practice we need patterns: *substitution* (undo a chain), *integration by parts* (trade derivative for product), *trigonometric identities* (reshape powers), and *partial fractions* (resolve rationals). Each technique is a controlled transformation: hypotheses act as guards, and the remaining difficulty decreases in a predictable way, useful for CDC style accounting (e.g., each parts step reduces polynomial degree by 1: a token style certificate).

18.1 Patterns and Rules

Lemma 18.1 (Substitution (indefinite/definite)). *If $u = g(x)$ with g differentiable and f continuous, then*

$$\int f(g(x)) g'(x) dx = \int f(u) du.$$

For definite integrals, if $x = a \rightarrow u = g(a)$ and $x = b \rightarrow u = g(b)$,

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

Lemma 18.2 (Integration by parts). *If u, v are differentiable, then*

$$\int u dv = uv - \int v du.$$

(LIATE heuristic: choose u from Logarithmic $\rightarrow$ Inverse trig $\rightarrow$ Algebraic $\rightarrow$ Trig $\rightarrow$ Exponential.)

Lemma 18.3 (Power-trig reshaping). (a) *If the power of $\sin$ or $\cos$ is odd, save one factor and use $\sin^2 x = 1 - \cos^2 x$ or $\cos^2 x = 1 - \sin^2 x$.*

(b) *If both powers are even, use half-angle: $\sin^2 x = \frac{1 - \cos 2x}{2}$, $\cos^2 x = \frac{1 + \cos 2x}{2}$.*

Lemma 18.4 (Partial fractions (proper rational p/q)). *If $\deg p < \deg q$ and q factors over $\mathbb{R}$ as*

$$q(x) = \prod_i (x - a_i)^{m_i} \prod_j (x^2 + b_j x + c_j)^{n_j} \quad (c_j > \frac{b_j^2}{4}),$$

then

$$\frac{p(x)}{q(x)} = \sum_i \sum_{k=1}^{m_i} \frac{A_{i,k}}{(x - a_i)^k} + \sum_j \sum_{k=1}^{n_j} \frac{B_{j,k}x + C_{j,k}}{(x^2 + b_j x + c_j)^k}.$$

If $\deg p \geq \deg q$ first do long division.

18.2 Worked Example A (definite substitution)

Example 18.5 ($\int_0^1 \frac{2x}{x^2+1} dx$). Let $u = x^2 + 1$, $du = 2x dx$. When $x = 0$, $u = 1$; when $x = 1$, $u = 2$.

$$\int_0^1 \frac{2x}{x^2+1} dx = \int_1^2 \frac{1}{u} du = [\ln u]_1^2 = \boxed{\ln 2}.$$

Guard. $u > 0$ on $[0, 1]$ so $\ln u$ is valid.

18.3 Worked Example B (integration by parts; degree drop)

Example 18.6 ($\int x e^{2x} dx$). Choose $u = x$ (algebraic), $dv = e^{2x} dx$. Then $du = dx$, $v = \frac{1}{2}e^{2x}$.

$$\int x e^{2x} dx = x \cdot \frac{1}{2}e^{2x} - \int \frac{1}{2}e^{2x} dx = \frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C = \boxed{\frac{e^{2x}}{4}(2x - 1) + C}.$$

CDC view. With $\Phi =$ polynomial degree, one parts step reduces Φ from 1 to 0 (token drop by 1); repeated parts on $\int x^n e^{ax}$ terminates in $n+1$ steps.

18.4 Worked Example C (trig powers)

Example 18.7 ($\int \sin^3 x \cos^2 x dx$). Save one $\sin x$: $\sin^3 x = (1 - \cos^2 x) \sin x$. Let $u = \cos x$, $du = -\sin x dx$.

$$\int \sin^3 x \cos^2 x dx = \int (1 - u^2)u^2(-du) = \int (u^2 - u^4) du = \frac{u^3}{3} - \frac{u^5}{5} + C = \boxed{\frac{\cos^3 x}{3} - \frac{\cos^5 x}{5} + C}.$$

18.5 Worked Example D (partial fractions: linear factors)

Example 18.8 ($\int \frac{x+2}{x^2-1} dx$). Factor: $x^2 - 1 = (x-1)(x+1)$. Decompose:

$$\frac{x+2}{x^2-1} = \frac{A}{x-1} + \frac{B}{x+1} \Rightarrow x+2 = A(x+1) + B(x-1).$$

Match coefficients: $A+B=1$, $A-B=2 \Rightarrow A=\frac{3}{2}$, $B=-\frac{1}{2}$. Thus

$$\int \frac{x+2}{x^2-1} dx = \frac{3}{2} \int \frac{dx}{x-1} - \frac{1}{2} \int \frac{dx}{x+1} = \boxed{\frac{3}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C}.$$

Guard. Exclude $x = \pm 1$ (domain of the integrand).

18.6 Worked Example E (irreducible quadratic)

Example 18.9 ($\int \frac{2x+1}{x^2+4x+5} dx$). Complete the square: $x^2 + 4x + 5 = (x+2)^2 + 1$. Write numerator as derivative + constant: $2x+1 = 2(x+2) - 3$.

$$\int \frac{2(x+2)}{(x+2)^2+1} dx - 3 \int \frac{dx}{(x+2)^2+1} = \ln((x+2)^2+1) - 3 \arctan(x+2) + C.$$

So

$$\boxed{\ln(x^2+4x+5) - 3 \arctan(x+2) + C}.$$

Key Takeaways

- **Substitution** inverts a chain: match $g'(x) dx$ with du and change limits for definite integrals.
- **By parts** trades complexity: choose u to make du simpler (degree drops for polynomials).
- **Trig powers** rely on parity and identities ($\sin^2 + \cos^2 = 1$, half-angles).
- **Partial fractions** reduce rationals to logs and arctans after division/expansion.

Try It Now

1) **Substitution (definite).** $\int_0^{\pi/2} \sin(3x) \cos(3x) dx.$

Answer: Let $u = \sin(3x)$, $du = 3 \cos(3x) dx$: $\frac{1}{3} \int_0^1 u du = \frac{1}{6}.$

2) **By parts.** $\int x^2 e^x dx.$

Answer (tabular): $e^x(x^2 - 2x + 2) + C.$

3) **Trig powers.** $\int \cos^4 x dx.$

Answer: Use $\cos^2 x = \frac{1+\cos 2x}{2}$: $\int \left(\frac{1+\cos 2x}{2}\right)^2 dx = \frac{3}{8}x + \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + C.$

4) **Partial fractions.** $\int \frac{3x+5}{(x-1)(x+2)} dx.$

Answer: Decompose to $A/(x-1) + B/(x+2)$ by solving $3x+5 = A(x+2) + B(x-1) \Rightarrow A = \frac{8}{3}, B = \frac{1}{3}.$ Thus

$$\int \frac{3x+5}{(x-1)(x+2)} dx = \frac{8}{3} \ln|x-1| + \frac{1}{3} \ln|x+2| + C.$$

On-Page Reference

Substitution: $u = g(x)$, $du = g'(x) dx \Rightarrow \int f(g)g' = \int f(u) du$; change limits for definite cases.

Parts: $\int u dv = uv - \int v du$; LIATE helps choose u . Tabular parts for repeated patterns.

Trig identities: $\sin^2 x = 1 - \cos^2 x$, $\cos^2 x = 1 - \sin^2 x$; $\sin^2 x = \frac{1-\cos 2x}{2}$, $\cos^2 x = \frac{1+\cos 2x}{2}$.

Partial fractions: divide if needed; expand over linear and irreducible quadratic factors; integrate to logs/arctans.

Squares: $x^2 + bx + c = (x + b/2)^2 + (c - b^2/4)$ (for arctan forms).

Alignment Notes

Assumptions: Domains respected (e.g., $\ln$ on > 0); denominators nonzero; continuity where FTC applies. **Composes with:** rational/piecewise analysis (breakpoints), optimization (work/energy), DEs (integrating factors). **CDC connector:** For $\int x^n e^{ax} dx$, let $\blacklozenge \Phi =$ degree n . Each parts step decreases Φ by 1 (token-style certificate) until termination. For definite integrals, substitution with new limits keeps the computation self-contained and auditable.

19. Series and Error Control

Context & Necessity

Power series turn complicated functions into polynomials with controllable error. Taylor polynomials provide local models; remainder bounds make the approximation auditable. For CDC, the *remainder* R_N is a natural potential $\diamond \Phi$: each extra term either reduces $\diamond \Phi$ by a predictable *token* (absolute bound) or by a *factor* (geometric/alternating tails), giving explicit N for any target ε .

19.1 Power Series and Convergence

Definition 19.1 (Power series about a). A power series has the form $\sum_{n=0}^{\infty} c_n(x-a)^n$. The set of x for which it converges is an interval centered at a with *radius of convergence* $R \in [0, \infty]$.

Lemma 19.2 (Ratio/root tests $\Rightarrow$ radius R). If $L = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|$ exists, then $R = \frac{1}{L}$ (with $1/0 = \infty$, $1/\infty = 0$). More generally (Cauchy–Hadamard), $\frac{1}{R} = \limsup_{n \rightarrow \infty} |c_n|^{1/n}$. For $|x-a| < R$ the series converges absolutely; for $|x-a| > R$ it diverges; endpoints require separate tests.

Lemma 19.3 (Termwise differentiation/integration). Inside $|x-a| < R$, the series can be differentiated or integrated termwise, and the resulting series have the same radius R .

19.2 Taylor Polynomials and Remainders

Definition 19.4 (Taylor polynomial). If f has n continuous derivatives near a , the degree- n Taylor polynomial is

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

The *remainder* is $R_n(x) = f(x) - T_n(x)$.

Theorem 19.5 (Taylor’s Theorem (Lagrange form)). If f has $n+1$ continuous derivatives on an interval containing a and x , then

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1} \quad \text{for some } \xi \text{ between } a \text{ and } x.$$

Hence, if $|f^{(n+1)}(t)| \leq M$ on the segment linking a and x ,

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}.$$

19.3 Canonical Maclaurin Series

For x in each domain:

$$\begin{aligned}
 e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad (R = \infty) & \sin x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \quad (R = \infty) \\
 \cos x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \quad (R = \infty) & \ln(1+x) &= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \quad (|x| < 1; x = -1 \text{ diverges}) \\
 \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n \quad (|x| < 1) & \arctan x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \quad (|x| < 1; x = \pm 1 \text{ conditionally})
 \end{aligned}$$

19.4 Worked Example A (Taylor bound for $e^{0.5}$)

Example 19.6 (Choose n for $|R_n| < 10^{-4}$ and evaluate T_n). Approximate $e^{0.5}$ with a Maclaurin polynomial. For $f^{(k)}(x) = e^x$, on $[0, 0.5]$ we have $|f^{(n+1)}(x)| \leq e^{0.5}$. The Lagrange bound gives

$$|R_n(0.5)| \leq \frac{e^{0.5}}{(n+1)!} (0.5)^{n+1}.$$

Smallest n with the bound $\leq 10^{-4}$ is $n = 5$ (since $\frac{e^{0.5}}{6!} 0.5^6 \approx 3.58 \times 10^{-5}$). Thus

$$T_5(0.5) = \sum_{k=0}^5 \frac{0.5^k}{k!} = 1.6486979167 \quad \text{and} \quad e^{0.5} \approx 1.6487212707,$$

with true error $\approx 2.34 \times 10^{-5}$ ($< 10^{-4}$).

19.5 Worked Example B (Alternating series error for $\sin x$)

Lemma 19.7 (Alternating Series Test with error). *If (a_n) is decreasing to 0 and $S = \sum (-1)^n a_n$, then the error of the N th partial sum satisfies $|S - S_N| \leq a_{N+1}$ and has the sign of the first omitted term.*

Example 19.8 (Two-term $\sin x$ at $x = 0.3$). $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$. With $x = 0.3$,

$$S_1 = x - \frac{x^3}{6} = 0.2955, \quad \text{error bound} \leq \frac{x^5}{5!} \approx 2.025 \times 10^{-5}.$$

Actual error $\approx 2.02 \times 10^{-5}$, within the bound.

19.6 Worked Example C (Log approximation with target tolerance)

Example 19.9 (Approximate $\ln(1.5)$ to $< 10^{-5}$ using $\ln(1+x)$). Take $x = 0.5$ in $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$ (alternating, decreasing). Choose N with $\frac{x^{N+1}}{N+1} < 10^{-5}$; $N = 12$ works since $0.5^{13}/13 \approx 9.39 \times 10^{-6}$. Hence

$$\ln(1.5) \approx \sum_{n=1}^{12} (-1)^{n+1} \frac{0.5^n}{n} = 0.40545869197,$$

true value 0.40546510810; error $\approx 6.42 \times 10^{-6}$ ($< 10^{-5}$), as guaranteed.

19.7 Operational Patterns

- **Pick the center a to minimize $|x - a|$.** Smaller $|x - a|$ tightens $|R_n|$ quickly.
- **Use Lagrange bounds when derivatives are monotone/bounded.** Supply $M \geq \max |f^{(n+1)}|$ on the segment.
- **Exploit alternating tails.** For alternating, decreasing terms, the first omitted term bounds the error.
- **Prefer geometric control when available.** If $|a_{n+1}/a_n| \leq r < 1$ eventually, each term multiplies the tail by $\lesssim r$ (factor-style).

Key Takeaways

- Radius R sets the region where series behave well; inside it, differentiate/integrate termwise.
- Taylor's Theorem gives a computable $|R_n|$; alternating/geometric structures give even sharper one-line bounds.
- Treat $|R_n|$ as $\blacklozenge \Phi$: choose n so $\blacklozenge \Phi \leq \varepsilon$, or add terms until the first omitted term drops below ε .

Try It Now

1) Cosine accuracy. Find the smallest n such that $|R_{2n}(0.2)| < 10^{-6}$ for $\cos x$ at $a = 0$.

Answer: $|R_{2n}(x)| \leq \frac{|x|^{2n+2}}{(2n+2)!}$.

Test $n = 1$: $\frac{0.2^4}{4!} = \frac{0.0016}{24} \approx 6.67 \times 10^{-5} > 10^{-6}$ (fails).

Test $n = 2$: $\frac{0.2^6}{6!} = \frac{0.000064}{720} \approx 8.89 \times 10^{-8} < 10^{-6}$ (passes).

Therefore the *smallest* n is $n = 2$, i.e., degree 4 suffices.

2) Geometric tail. Approximate $\frac{1}{1-x}$ at $x = 0.1$ with $\sum_{k=0}^N x^k$. Choose N for tail $< 10^{-8}$.

Answer: Tail $= \frac{x^{N+1}}{1-x} = \frac{0.1^{N+1}}{0.9} < 10^{-8} \Rightarrow 0.1^{N+1} < 9 \times 10^{-9} \Rightarrow N+1 \geq 9$; take $N = 8$.

3) Leibniz for $\arctan(1)$. Choose N so the Leibniz series $\arctan 1 = \sum_{n=0}^{\infty} (-1)^n \frac{1}{2n+1}$ is within 10^{-5} .

Answer: First omitted term bound $\frac{1}{2N+3} \leq 10^{-5} \Rightarrow N \geq 49999$ (slow convergence!).

4) Differentiate a series. Use termwise differentiation to show $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}$ for $|x| < 1$.

Answer: Differentiate $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$.

On-Page Reference

Radius: $R = 1/\limsup |c_n|^{1/n}$; ratio test when limit exists.

Taylor remainder: $|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}$ with $M \geq \max |f^{(n+1)}|$ on $[a, x]$.

Alternating error: $|S - S_N| \leq$ first omitted term (if decreasing to 0).

Geometric tail: $\frac{r^{N+1}}{1-r}$ for $|r| < 1$.

Series table: e^x , $\sin x$, $\cos x$, $\ln(1+x)$, $(1-x)^{-1}$, $\arctan x$ with domains as listed above.

Alignment Notes

Assumptions: Work inside the radius R ; check endpoint behavior separately. Bounds use monotonicity of derivatives or alternating-term decay where stated. **Composes with:** linearization (Act III), integration techniques (termwise), limits (precise). **CDC connector:** Treat the remainder bound $|R_N|$ as $\blacklozenge \Phi$. *Token style:* Lagrange bound shrinks superlinearly with N (factorial in the denominator). *Factor style:* geometric/alternating tails shrink by a fixed factor per step, giving explicit $N \geq \lceil \log(\Phi_0/\varepsilon)/\log(1/r) \rceil$ when $|a_{n+1}/a_n| \leq r < 1$.

Part V

Act V — CDC Toolkit

20. CDC Step 2: Choice, Parallel, and Invariants

Context & Necessity

Real systems have many legal moves at once. *Choice* (nondeterminism), *parallel updates*, and *invariants* (always true assertions) give a disciplined way to reason safely and progress even when multiple rules compete. The CDC aim: retain stepwise ✓ **C** certificates while ensuring steps do not interfere.

20.1 Operational Forms

Definition 20.1 (Nondeterministic choice). Given rules $U_1, \dots, U_m$ with guards G_i , a *choice step* picks any i with $G_i(x)$ true and applies $x \leftarrow U_i(x)$.

Definition 20.2 (Parallel batch). A *parallel step* chooses an index set $S \subseteq \{1, \dots, m\}$ and applies all U_i for $i \in S$ in one logical tick. We require either (i) *disjoint writes* (no two rules update the same component), or (ii) *commutativity*: $U_i \circ U_j = U_j \circ U_i$ on states where both guards hold.

Definition 20.3 (Invariant). A predicate $I : X \rightarrow \{\text{T}, \text{F}\}$ is an *invariant* for a rule set if $I(x_0)$ holds initially and $I(x) \wedge G_i(x) \Rightarrow I(U_i(x))$ for all i .

Theorem 20.4 (Safety via invariants). *If I is an invariant and $I(x_0)$, then I holds after any finite sequence of choice/parallel steps.*

Lemma 20.5 (Progress under choice (sum and lexicographic potentials)). *Let $\Phi = \sum_{k=1}^K \Phi_k \geq 0$. If each rule U_i that changes component k satisfies a token drop $\Phi'_k \leq \Phi_k - \delta_i$ with $\delta_i > 0$, then any sequence of enabled steps yields Φ strictly decreasing until all relevant guards are false. For staged tasks, use a lexicographic potential $(\Phi_1, \Phi_2, \dots)$: first decrease Φ_1 to 0, then Φ_2 , etc.*

20.2 Worked Example A (even–odd transposition sort)

Example 20.6 (Parallel adjacent swaps without interference). Array $a[1..n]$. Two phases:

even phase: swap $(2, 3), (4, 5), \dots$; odd phase: swap $(1, 2), (3, 4), \dots$,

whenever a pair is inverted.

◆ $\Phi = \text{Inv}(a)$ (number of inversions). G any inverted adjacent pair in the chosen phase. $\rightarrow$ **U** swap all inverted pairs of that phase in *parallel* (disjoint writes). ✓ **C** each swap removes at least one inversion; disjoint pairs sum their drops. □ **H** no adjacent inversions $\Rightarrow$ array sorted.

Invariant. Array elements are a permutation of the input (preserved by swaps).

Progress. Each phase decreases Φ by at least the number of swapped pairs; hence after $O(n^2)$ swaps (and $O(n)$ phases) $\Phi = 0$.

20.3 Worked Example B (coordinate descent without conflicts)

Example 20.7 (Two-coordinate descent with safe steps). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be differentiable with partial Lipschitz constants L_x, L_y :

$$|\partial_x f(x + \Delta, y) - \partial_x f(x, y)| \leq L_x |\Delta|, \quad |\partial_y f(x, y + \Delta) - \partial_y f(x, y)| \leq L_y |\Delta|.$$

Define rules

$$U_x : (x, y) \mapsto (x - \frac{1}{L_x} \partial_x f(x, y), y), \quad U_y : (x, y) \mapsto (x, y - \frac{1}{L_y} \partial_y f(x, y)).$$

◆ $\Phi = f$. **G** always. ✓ **C** (Descent Lemma, coordinate form) gives

$$f(U_x(x, y)) \leq f(x, y) - \frac{1}{2L_x} (\partial_x f)^2, \quad f(U_y(x, y)) \leq f(x, y) - \frac{1}{2L_y} (\partial_y f)^2.$$

Under *alternating choice* or *even-odd scheduling*, Φ drops each step.

20.4 Worked Example C (prefix-balance invariant)

Example 20.8 (Parentheses scan). Scan a string of '(' and ')'. Maintain counter c (increment for '(', decrement for ')'). Invariant I : *every prefix has $c \geq 0$* . The rule that consumes the next symbol preserves I . At the end $c = 0$ certifies a well-formed string; $c < 0$ at any point violates I immediately.

Key Takeaways

- Invariants prove *safety*; potentials prove *progress*. Use both.
- Parallel batches need disjoint writes or commuting updates; then token drops add.
- Sum and lexicographic potentials manage multi-goal progress under nondeterminism.

Try It Now

- 1) Invariant.** Show that swapping adjacent entries preserves multiset of values.
Hint: Swaps are permutations.
- 2) Parallel safety.** For even-odd transposition, argue that pairs in a phase are disjoint.
Hint: Indices differ by at least 2.
- 3) Lexicographic.** Design (Φ_1, Φ_2) so that Φ_1 counts out-of-place max elements (left of their final positions) and Φ_2 counts inversions among the rest.

On-Page Reference

Invariant: preserved by every enabled rule. **Parallel:** disjoint writes or commutativity.
Potentials: sum and lexicographic forms for multi-stage progress.

Alignment Notes

Assumptions: deterministic arithmetic; rules defined where guards hold. **Composes with:** Step 1 token/factor lemmas; later scheduling/fairness. **CDC connector:** certify per-rule drops and ensure rules do not interfere.

21. CDC Step 3: Step Size, Smoothness, and Certified Descent

Context & Necessity

To turn $\Phi = f$ into guaranteed decrease in calculus-based updates, we need *smoothness* assumptions and *safe step sizes*. This chapter records descent lemmas that provide one-line ✓ C certificates for gradient-like moves and their discrete variants.

21.1 Smoothness and Curvature

Definition 21.1 (Lipschitz gradient (L -smooth)). $f : \mathbb{R}^d \rightarrow \mathbb{R}$ has L -Lipschitz gradient if $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$.

Lemma 21.2 (Descent Lemma). *If f is L -smooth, then for all x and $\eta > 0$,*

$$f(x - \eta \nabla f(x)) \leq f(x) - \left(\eta - \frac{L\eta^2}{2}\right) \|\nabla f(x)\|^2.$$

Hence any $\eta \in (0, 2/L]$ yields a token drop; in particular $\eta = 1/L$ gives $f(x^+) \leq f(x) - \frac{1}{2L} \|\nabla f(x)\|^2$.

Definition 21.3 (Strong convexity). f is μ -strongly convex if $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2} \|y - x\|^2$ for all x, y .

Lemma 21.4 (Linear convergence under μ - L). *If f is μ -strongly convex and L -smooth, then with $\eta = 1/L$,*

$$f(x_{k+1}) - f^* \leq \left(1 - \frac{\mu}{L}\right) (f(x_k) - f^*).$$

Thus $\blacklozenge \Phi = f - f^*$ yields factor certificate with $c = 1 - \mu/L$.

21.2 Backtracking (Armijo) as a certificate engine

Definition 21.5 (Armijo condition). Given $\alpha \in (0, 1)$, $\beta \in (0, 1)$, choose the smallest $m \geq 0$ with

$$f(x - \beta^m \eta_0 \nabla f(x)) \leq f(x) - \alpha \beta^m \eta_0 \|\nabla f(x)\|^2.$$

Then set $\eta = \beta^m \eta_0$.

This picks a step that *proves* a token drop $\alpha \eta \|\nabla f\|^2$ without knowing L .

21.3 Worked Example A (quadratic)

Example 21.6 ($f(x) = \frac{1}{2}x^\top Qx + b^\top x$ with $0 \prec \mu I \preceq Q \preceq LI$). With $\eta = 1/L$,

$$f(x^+) - f^* \leq \left(1 - \frac{\mu}{L}\right)(f(x) - f^*).$$

Hence $k \geq \lceil \log_{1/(1-\mu/L)}(\frac{f(x_0)-f^*}{\varepsilon}) \rceil$ steps achieve $f - f^* \leq \varepsilon$.

21.4 Worked Example B (backtracking on a 1D function)

Example 21.7 ($f(x) = e^x - x$ at $x_0 = 0$). $\nabla f(0) = 1$. With $\eta_0 = 1$, $\alpha = \frac{1}{4}$, $\beta = \frac{1}{2}$: check $m = 0$: $f(-1) = e^{-1} + 1 \approx 1.3679 > 0.75$ fails the Armijo RHS = 0.75. Try $m = 1$ ($\eta = \frac{1}{2}$): $f(-1/2) = e^{-1/2} + 1/2 \approx 1.1065 \leq 1 - 0.25 \cdot 0.5 = 0.875$ fails. Try $m = 2$ ($\eta = \frac{1}{4}$): $f(-1/4) \approx 0.8866 \leq 0.9375$ holds. Certified token drop $\alpha\eta\|\nabla f\|^2 = 1/16$.

21.5 Worked Example C (projected step)

Example 21.8 (Projected gradient). On a closed convex set $\mathcal{C}$, update $x^+ = \Pi_{\mathcal{C}}(x - \eta\nabla f(x))$. Nonexpansiveness of $\Pi_{\mathcal{C}}$ and L -smoothness give the same Armijo/backtracking logic, so we retain a token drop for small enough η .

Key Takeaways

- L -smooth + small $\eta \Rightarrow$ one-line token certificate; μ -strongly convex gives factor rates.
- Backtracking finds a provably safe η without L .
- Projections preserve descent guarantees on convex sets.

Try It Now

1) Safe window. For $L = 10$, which η guarantee descent by the Descent Lemma?

Answer: $\eta \in (0, 0.2]$.

2) Rate. With $\mu/L = 0.1$, how many steps to reduce $f - f^*$ by 10^{-3} ?

Answer: Need $k \geq \lceil \log(10^{-3}) / \log(0.9) \rceil \approx 66$.

3) Armijo. Show that for L -smooth f , any $\eta \leq \frac{1}{L}$ satisfies Armijo with $\alpha = \frac{1}{2}$.

On-Page Reference

Smoothness: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$. **Descent:** $f(x - \eta g) \leq f(x) - (\eta - \frac{L\eta^2}{2})\|g\|^2$.

Strong convexity: factor rate $1 - \mu/L$. **Armijo:** choose η to certify a token drop without L .

Alignment Notes

Assumptions: differentiability; convexity as stated for factor rates. **Composes with:** choice/parallel (coordinate descent), scheduling/fairness. **CDC connector:** $\diamond \Phi = f - f^*$; token/factor certificates flow from smoothness.

22. CDC Schedules, Fairness, and Amortized Progress

Context & Necessity

When many rules compete, *who moves when* matters. Schedules (round-robin, random, adversarial) interact with guarantees. Fairness assumptions ensure that perpetually enabled rules eventually fire. Amortized analysis certifies overall progress when individual steps sometimes regress but the average improves.

22.1 Schedules and Fairness

Definition 22.1 (Schedule). A schedule is an infinite (or finite) sequence of rule indices $i_0, i_1, \dots$.

Definition 22.2 (Weak fairness). If a guard G_i is true at all but finitely many times, then i occurs infinitely often in the schedule.

Theorem 22.3 (Progress under fairness with token drops). *Let $\Phi = \sum_k \Phi_k \geq 0$. Suppose each rule i that touches component k guarantees $\Phi'_k \leq \Phi_k - \delta_i$ when G_i holds. Under any weakly fair schedule, eventually all such guards become false; if halting is defined by G_i all false, the process halts in finitely many steps.*

22.2 Amortized Certificates

Definition 22.4 (Banker method). Augment Φ with a nonnegative *bank* B . Assign credits on some steps and spend them on others. If $\Phi + B$ decreases by at least $\delta > 0$ each step, the total number of steps is $\leq \lceil (\Phi_0 + B_0)/\delta \rceil$ even if Φ occasionally increases.

Lemma 22.5 (Round-robin aggregate drops). *If a cycle of m rules ensures $\sum_{t=1}^m \Delta\Phi_t \leq -\Delta$ every cycle, then after q cycles $\Phi \leq \Phi_0 - q\Delta$.*

22.3 Worked Example A (coordinate descent under round-robin)

Example 22.6 (Cycle-wise factor). Let f be μ -strongly convex, L -smooth with coordinates $(x_1, \dots, x_m)$. Update in cycles $1, \dots, m$ with steps $\eta_j = 1/L_j$. There exist constants $c_j < 1$ so that f after one full cycle satisfies $f_{k+m} - f^* \leq c(f_k - f^*)$ with $c = \prod_j c_j$. Thus $q \geq \lceil \log(\frac{f_0 - f^*}{\epsilon}) / \log(1/c) \rceil$ cycles suffice.

22.4 Worked Example B (even–odd transposition: time to sort)

Example 22.7 (Phase bound). Let $W = \sum_{i < j} \mathbf{1}(a_i > a_j)$ be inversions. In each phase (even or odd) at least one inversion is removed whenever $W > 0$. Moreover an element can move at most one position per phase, so after at most n phases the maximum index error decreases by at least 1. Combining: $\leq n$ phases suffice to eliminate all inversions, giving an $O(n^2)$ swap bound.

22.5 Worked Example C (amortized queue with two stacks; sketch)

Example 22.8 (Push/Pop costs). Maintain two stacks (in, out); push to in; pop from out or transfer all of in to out when needed. Potential $\Phi = \text{size of } in$. Each push increases Φ by 1; a transfer of k items decreases Φ by k and pays for k moves. Amortized cost per operation is $O(1)$ though individual transfers cost $O(k)$.

Key Takeaways

- Fairness converts local token drops into global termination.
- Amortized potentials (*banker* method) handle temporary regressions.
- Cycle-level reasoning yields factor bounds across repeated schedules.

Try It Now

- 1) Weak fairness.** Give a schedule that is unfair to a perpetually enabled rule.
Answer: Always pick rule 1; rule 2 never fires though G_2 holds.
- 2) Banker design.** For a data structure where occasional rehash costs $O(n)$, choose $\Phi + B$ so that average cost is $O(1)$.
Hint: Bank grows by $O(1)$ per op and pays for rehash.
- 3) Cycle factor.** If three rules yield per-step factors c_1, c_2, c_3 , show a cycle factor $c = c_1 c_2 c_3$.

On-Page Reference

Fairness: enabled infinitely often $\Rightarrow$ scheduled infinitely often. **Amortized:** augment with bank to smooth spikes. **Cycle bounds:** multiply factors or add token drops per cycle.

Alignment Notes

Assumptions: guards observable; scheduler model fixed. **Composes with:** Step 2 (invariants), Step 3 (descent). **CDC connector:** scheduling hypotheses turn per-step certificates into end-to-end guarantees.

23. CDC Templates, Checklists, and Capstones

Context & Necessity

Repeatable proof obligations keep CDC work auditable. This chapter provides fill-in templates, common certificate patterns, and two capstones you can adapt: bisection (discrete) and gradient descent (continuous).

23.1 CDC Proof Sheet (template)

Name of process:	
State x :	
Potential $\blacklozenge \Phi \geq 0$:	
Guards $\textcolor{red}{G}$:	
Update(s) $\rightarrow \textcolor{blue}{U}$:	
Certificate(s) $\checkmark \textcolor{teal}{C}$:	
Halt $\square \textcolor{teal}{H}$:	
Invariant(s):	
Parallel safety (disjoint/commute):	
Schedule/fairness assumption:	
Global bound (tokens/factors/lexicographic/banker):	

23.2 Certificate Patterns

- **Token style:** $\Phi' \leq \Phi - \delta$ with static $\delta > 0$ or data-driven $\delta(x)$ bounded below.
- **Factor style:** $\Phi' \leq c\Phi$ ($0 < c < 1$), sometimes per cycle.
- **Affine mix:** $\Phi' \leq c\Phi - \delta$; iterate to get $\Phi_k \leq c^k \Phi_0 - \delta \sum_{i=0}^{k-1} c^i$.
- **Lexicographic:** $(\Phi_1, \Phi_2, \dots)$ with short-circuit comparison.
- **Amortized:** $\Phi + B$ decreases every step even if Φ spikes.

23.3 Capstone A (Bisection; end-to-end)

Example 23.1 (Root finding with explicit step bound). $\blacklozenge \Phi = R - L$; $\textcolor{red}{G} f(L)f(R) \leq 0$; $\rightarrow \textcolor{blue}{U}$ midpoint and keep sign-change half; $\checkmark \textcolor{teal}{C} \Phi' \leq \frac{1}{2}\Phi$; $\square \textcolor{teal}{H} \Phi \leq \varepsilon$. By composition: $k \geq \lceil \log_2(\Phi_0/\varepsilon) \rceil$ steps.

23.4 Capstone B (Gradient descent; end-to-end)

Example 23.2 (Smooth strongly convex). $\blacklozenge \Phi = f - f^*$; $\mathbf{G}$ always; $\rightarrow \mathbf{U} \ x \leftarrow x - \frac{1}{L} \nabla f(x)$; $\checkmark \mathbf{C}$
 $\Phi' \leq (1 - \mu/L)\Phi$; $\square \mathbf{H} \ \Phi \leq \varepsilon$. Bound: $k \geq \left\lceil \frac{\ln((f_0 - f^*)/\varepsilon)}{\ln(1/(1 - \mu/L))} \right\rceil$.

23.5 How to Debug a Failed Certificate

- Bad potential:** can Φ increase or go negative? Try absolute errors, counts, or energies.
- Guard gaps:** is there a state where no guard holds but the goal is unmet? Add a fallback rule.
- Interference:** do parallel updates touch the same location? Partition into disjoint batches.
- Missing constants:** choose step sizes from smoothness/Lipschitz or add backtracking.
- Scheduler assumptions:** require fairness or enforce round-robin.

Key Takeaways

- A CDC proof is a short checklist: *state, potential, guards, updates, certificates, halt, invariants, schedule*.
- Token/factor/lexicographic/banker cover most practical progress shapes.
- Parallel safety reduces to disjoint updates or commutativity; otherwise batch.

Try It Now

1) Affine mix. Suppose $\Phi' \leq 0.9\Phi - 3$ until $\Phi < 30$. Give a bound on steps from $\Phi_0 = 120$ to $\Phi < 30$.

Answer: Iterate $\Phi_k \leq 0.9^k \cdot 120 - 3 \frac{1-0.9^k}{1-0.9}$. Solve $0.9^k \cdot 120 - 30(1 - 0.9^k) < 30 \Rightarrow 0.9^k < 0.25 \Rightarrow k > \log_{1/0.9}(4) \approx 13.2$; take 14.

2) Invariant find. For Dijkstra's algorithm, propose an invariant about the settled set S .
Answer (sketch): Each $v \in S$ has final shortest distance; edges out of S respect the frontier bound.

3) Lexicographic sketch. For a two-phase cleanup, define Φ_1 as remaining phase-1 tasks, Φ_2 as remaining phase-2 tasks; show termination.

On-Page Reference

Patterns: token, factor, affine, lexicographic, amortized. **Proof sheet:** concise obligations. **Capstones:** bisection and gradient descent with explicit step counts.

Alignment Notes

Assumptions: standard smoothness/convexity where invoked. **Composes with:** all prior acts; serves as a reference for new CDC constructions. **CDC connector:** This chapter is the audit toolkit use it as a reference to certify any new processes you introduce.

Why Conditional Descent Calculus (CDC): A One-Page Rationale

Philosophy: Auditable Progress

Two ancient questions drive CDC: *What does it mean to make progress?* and *How can we know we did?* CDC answers with small, checkable obligations at the level of a single step: choose a nonnegative $\blacklozenge \Phi$ (a potential to decrease), specify guards $\mathbf{G}$ (when a rule may act), an update $\rightarrow \mathbf{U}$ (what changes), and a one-line certificate $\checkmark \mathbf{C}$ that $\blacklozenge \Phi$ strictly drops (token: $\Phi' \leq \Phi - \delta$) or shrinks by a factor (factor: $\Phi' \leq c\Phi$). Because these obligations are local, they are *auditable*: a reader can verify them without rederiving the whole method. Because they compose (drops add; factors multiply; affine mixes combine), they scale to complex systems.

Mathematics: Potentials, Invariants, and Composition

CDC isolates the algebra behind termination and convergence.

- **Potentials** $\Phi \geq 0$ formalize “getting closer” (size of a set, error to a fixed point, energy).
- **Invariants** keep us safe: predicates I preserved by every enabled step.
- **Composition** is algebraic: token drops add ($\Delta\Phi \geq \delta_A + \delta_B$); factor steps multiply (c_{ACB}); mixed steps yield affine bounds ($\Phi' \leq c\Phi - \delta$).
- **Scheduling/parallelism** enter cleanly: disjoint/commuting updates let token drops sum across a batch; fairness turns per-step guarantees into end-to-end termination.

CDC is not a new theorem; it is a *discipline*: write your process as tiles, certify locally, compose globally.

Science: From Models to Guarantees

In science and engineering we measure, model, and iterate. CDC closes the loop with guarantees:

- **Discrete search**: bisection’s halving certificate gives exact step counts $k \geq \lceil \log_2(\text{width}_0/\varepsilon) \rceil$.
- **Dynamics and control**: Lyapunov functions certify stability: $\Phi(t + \Delta) \leq e^{-k\Delta}\Phi(t)$.
- **Optimization**: smoothness (L) and curvature (μ) give provable descent: $f(x^+) \leq f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$ (token) or $f_k - f^* \leq (1 - \mu/L)^k(f_0 - f^*)$ (factor).
- **Parallel computation**: under disjoint writes, many small decreases add.

CDC’s value is practical: it turns “works in practice” into *audited progress with a budget*. Choose $\blacklozenge \Phi$, state $\mathbf{G}/\rightarrow \mathbf{U}$, prove $\checkmark \mathbf{C}$, define $\square \mathbf{H}$ (when to stop). Repeat.

A. Notation, Symbols, and Tile Legend

CDC Tiles

$\blacklozenge \Phi$	Potential (nonnegative function measuring progress)
$\textcolor{red}{G}$	Guard (predicate enabling a rule)
$\rightarrow \textcolor{blue}{U}$	Update (state transition)
$\checkmark \textcolor{teal}{C}$	Certificate (local guarantee: token/factor/affine)
$\square \textcolor{black}{H}$	Halt condition (goal reached or no guard enabled)

General Notation

- $\mathbb{N} = \{0, 1, 2, \dots\}$, $\mathbb{R}$ real numbers; vectors are columns; $\|\cdot\|$ Euclidean norm.
- “Token” step: $\Phi' \leq \Phi - \delta$ with $\delta > 0$; “Factor” step: $\Phi' \leq c\Phi$ with $0 < c < 1$.
- Composition: token drops add; factor constants multiply; mixed: $\Phi' \leq c\Phi - \delta$.
- $\mathbf{1}(\cdot)$ is an indicator; $\lceil \cdot \rceil$ ceiling; log is natural unless base stated.

Function Regularity

- L -smooth (Lipschitz gradient): $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$.
- μ -strongly convex: $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2$.

B. Reference Tables

B.1 Trig and Exponential Basics

$$\sin^2 x + \cos^2 x = 1,$$

$$\sin(2x) = 2 \sin x \cos x,$$

$$a^x = e^{x \ln a},$$

$$\tan x = \frac{\sin x}{\cos x},$$

$$\cos(2x) = \cos^2 x - \sin^2 x.$$

$$\log_a x = \frac{\ln x}{\ln a}.$$

B.2 Derivative Table (domains implicit)

$$(x^n)' = nx^{n-1}$$

$$(\ln x)' = 1/x$$

$$(\sin x)' = \cos x$$

$$(\tan x)' = \sec^2 x$$

$$(e^x)' = e^x$$

$$(a^x)' = (\ln a)a^x$$

$$(\cos x)' = -\sin x$$

$$(\arctan x)' = 1/(1+x^2)$$

B.3 Integral Table

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \cos x dx = \sin x + C$$

B.4 Series (about 0)

$$e^x = \sum_{n \geq 0} \frac{x^n}{n!}$$

$$\cos x = \sum_{n \geq 0} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$\sin x = \sum_{n \geq 0} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\ln(1+x) = \sum_{n \geq 1} (-1)^{n+1} \frac{x^n}{n} \quad (|x| < 1)$$

B.5 Inequality Patterns

- AM-GM: for $a, b \geq 0$, $\frac{a+b}{2} \geq \sqrt{ab}$.

- Cauchy-Schwarz: $|\langle u, v \rangle| \leq \|u\| \|v\|$.

- For $|x| < 1$, $\frac{1}{1-x} = \sum_{n \geq 0} x^n$ gives geometric tail $r^{N+1}/(1-r)$.

C. CDC Mini-Cookbook

C.1 Twelve Canonical Patterns

1. **Bisection (root bracketing):** $\blacklozenge \Phi = R - L$; $\color{red}\mathbf{G} f(L)f(R) \leq 0$; $\rightarrow \color{blue}\mathbf{U}$ bisect keep sign-change; $\checkmark \color{teal}\mathbf{C} \Phi' \leq \frac{1}{2}\Phi$; $\square \color{black}\mathbf{H} \Phi \leq \varepsilon$.
2. **Binary search (discrete):** $\blacklozenge \Phi =$ candidate set size; $\color{red}\mathbf{G} \Phi > 1$; $\rightarrow \color{blue}\mathbf{U}$ keep consistent half; $\checkmark \color{teal}\mathbf{C}$ factor $1/2$; $\square \color{black}\mathbf{H} \Phi = 1$.
3. **Inversion sorting:** $\blacklozenge \Phi = \text{Inv}(a)$; $\color{red}\mathbf{G}$ adjacent inversion exists; $\rightarrow \color{blue}\mathbf{U}$ swap that pair; $\checkmark \color{teal}\mathbf{C}$ token $\delta = 1$; $\square \color{black}\mathbf{H}$ none remain.
4. **Euclid's algorithm:** $\blacklozenge \Phi = b$; $\color{red}\mathbf{G} b \neq 0$; $\rightarrow \color{blue}\mathbf{U} (a, b) \mapsto (b, a \bmod b)$; $\checkmark \color{teal}\mathbf{C}$ strict descent on $\mathbb{N}$; $\square \color{black}\mathbf{H} b = 0$.
5. **Newton cooling:** $\blacklozenge \Phi = |y - T_{\text{env}}|$; $\rightarrow \color{blue}\mathbf{U}$ advance Δt ; $\checkmark \color{teal}\mathbf{C}$ factor $e^{-k\Delta t}$.
6. **Gradient descent (L -smooth):** $\blacklozenge \Phi = f - f^*$; $\rightarrow \color{blue}\mathbf{U} x \leftarrow x - \frac{1}{L}\nabla f$; $\checkmark \color{teal}\mathbf{C}$ factor $1 - \mu/L$ if μ -strongly convex (token otherwise).
7. **Coordinate descent:** $\blacklozenge \Phi = f$; $\rightarrow \color{blue}\mathbf{U}$ per-coordinate $-\frac{1}{L_i}\partial_i f$; $\checkmark \color{teal}\mathbf{C}$ token per step; cycle factor.
8. **Projected step (constraints):** $\blacklozenge \Phi = f - f^*$; $\rightarrow \color{blue}\mathbf{U}$ project after gradient step; $\checkmark \color{teal}\mathbf{C}$ Armijo token.
9. **Amortized queue (two stacks):** $\blacklozenge \Phi = \text{size}(\text{in})$; $\color{red}\mathbf{G}$ op available; $\rightarrow \color{blue}\mathbf{U}$ push/pop/transfer; $\checkmark \color{teal}\mathbf{C}$ banker: $\Phi + B$ decreases by ≥ 1 per op.
10. **Even-odd transposition (parallel):** $\blacklozenge \Phi = \text{Inv}$; $\rightarrow \color{blue}\mathbf{U}$ disjoint swaps per phase; $\checkmark \color{teal}\mathbf{C}$ phase-wise token sum.
11. **Fixed-point iteration** $x \mapsto g(x)$: $\blacklozenge \Phi = |x - x^*|$; $\color{red}\mathbf{G}$ always; $\rightarrow \color{blue}\mathbf{U} x \leftarrow g(x)$; $\checkmark \color{teal}\mathbf{C}$ factor $|g'(x^*)|$ locally; $\square \color{black}\mathbf{H} \Phi \leq \varepsilon$.
12. **Mesh refinement (integration):** $\blacklozenge \Phi =$ mesh size Δ ; $\rightarrow \color{blue}\mathbf{U}$ halve Δ ; $\checkmark \color{teal}\mathbf{C}$ error $\propto \Delta^p$, factor 2^{-p} per refinement.

C.2 CDC Proof Sheet (blank)

State: _____
Potential: _____
Guards: _____
Update(s): _____
Certificate(s): _____
Halt: _____
Invariant(s): _____
Parallel safety: _____
Schedule/fairness: _____
Global bound: _____

C.3 Debugging Guide (fast)

- **Potential rises?** Switch to absolute error, counts, or energies; ensure non-negativity.
- **Stuck guard?** Add a fallback rule or widen the guard.
- **Interference?** Partition into disjoint batches or enforce a schedule.
- **No constants?** Use backtracking or estimate L, μ .

Index of Symbols

Symbol	Meaning
Sets & Numbers	
$\mathbb{R}$	Real numbers.
$\mathbb{N}, \mathbb{Z}$	Nonnegative integers; integers.
$[a, b], (a, b)$	Closed / open interval in $\mathbb{R}$.
Functions & Calculus	
$f'(x), f''(x)$	First and second derivative at x .
$\nabla f(x)$	Gradient (column of partials).
$\partial_i f(x)$	i th partial derivative at x .
$D^2 f(x), H_f(x)$	Hessian (matrix of second partials).
$\int_a^b f(x) dx$	Definite integral on $[a, b]$.
$F'(x) = f(x)$	F is an antiderivative of f .
Linear Algebra & Norms	
$\langle u, v \rangle$	Inner product (Euclidean unless stated).
$\ x\ $	Euclidean norm of a vector x .
$\ A\ $	Operator (spectral) norm of a matrix A .
I	Identity matrix (size by context).
$\lambda_{\min}(A), \lambda_{\max}(A)$	Smallest / largest eigenvalue of symmetric A .
Optimization & Rates	
f^*	Optimal value $\min_x f(x)$ (when defined).
$\arg \min_x f(x)$	Set of minimizers of f .
L	Lipschitz gradient constant (“ L -smooth”).
μ	Strong-convexity constant.
η	Step size / learning rate.
$c \in (0, 1)$	Factor in a factor-style certificate ($\Phi' \leq c \Phi$).
$\delta > 0$	Token drop in a token-style certificate ($\Phi' \leq \Phi - \delta$).
$\varepsilon > 0$	Tolerance / accuracy target.
Δ	Mesh size or generic small increment.

Symbol	Meaning
CDC Tiles & Meta	
$\blacklozenge \Phi$	Potential (nonnegative progress measure).
$\mathbf{G}$	Guard (predicate enabling an update).
$\rightarrow \mathbf{U}$	Update (state transition rule).
$\checkmark \mathbf{C}$	Certificate (local guarantee: token/factor/affine).
$\square \mathbf{H}$	Halt condition (goal reached / termination).
$I(\cdot)$	Invariant predicate (preserved by all enabled steps).
B	Bank (amortized-analysis credit).
S	Schedule (sequence of chosen rules).
Miscellany	
$\mathbf{1}(\cdot)$	Indicator function (1 if statement true, else 0).
$\text{sgn}(x)$	Sign of x ($-1, 0, +1$).
$\text{diag}(\cdot)$	Diagonal matrix with given entries.
$\det A$	Determinant of matrix A .
$\lceil t \rceil, \lfloor t \rfloor$	Ceiling / floor of real number t .
$\log, \ln$	Logarithm (base e unless base specified).

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